Supplement to *Testing for Treatment-Induced Recording Change with Few Clusters*

Ethan Xie

This supplement contains the proofs of every result in the paper, the simulation and application evidence behind the numbers the paper reports, the extended discussion of the related and applied literature, and the figure of exact size against the number of restrictions. Numbering of results follows the paper: a reference to Theorem 1 here is Theorem 1 there. Sections, tables and figures belonging to this supplement carry an S, so Table S1 is a table of this supplement and Table 1 is the table of the paper.

S1. PROOFS

Proof of Lemma 1. Write s_1 for the structure of the statement: latent outcomes Y^* satisfying parallel trends, recording paths m_c with $m_c(\ell) = 0$ for $\ell < 0$, and unit by category terms μ_{ic}, period by category terms ϕ_{tc}, recording noise u_{itc} and cohort labels G_i. By Assumption 1 the observed panel under s_1 is

$$Y_{itc} = Y^*_{itc} + \mu_{ic} + \phi_{tc} + m_c(\ell_{it})D_{it} + u_{itc}. \tag{S1.1}$$

Construct s_2 on the same probability space by leaving μ, ϕ, u and G unchanged, setting the recording paths to zero, and replacing the latent untreated potential outcome by

$$Y^\dagger_{itc}(\infty) = Y^*_{itc}(\infty) + m_c(t-g)\mathbf{1}\{t \geq g\} \qquad \text{on } \{G_i = g\}, \tag{S1.2}$$

with treated potential outcomes shifted by the same increment, so that latent treatment effects are unchanged. Since s_2 has no recording response, its observed panel is $Y_{itc} = Y^\dagger_{itc} + \mu_{ic} + \phi_{tc} + u_{itc}$, and substituting (S1.2) and $\ell_{it} = t - G_i$ returns (S1.1) cell by cell, pointwise on the underlying sample space. The two structures therefore induce the same joint law of $\{(Y_{itc})_{t,c}, G_i\}_{i \leq N}$, not merely the same moments.

For the deviation induced by s_2, note that s_1 satisfies latent parallel trends, so any departure under s_2 is the increment in (S1.2). Differencing it across t and across cohorts in the definition of $\Delta_c(g,t)$ gives $\Delta_c(g,t) = m_c(t-g)\mathbf{1}\{t \geq g\}$, which is (3.2). It vanishes for $t < g$ because $m_c(\ell) = 0$ for $\ell < 0$, which is the no anticipation restriction in the recording layer. □

Proof of Corollary 1. Let φ be measurable with respect to $\sigma(\{(Y_{itc})_{t,c}, G_i\}_{i \leq N})$. By Lemma 1 the law of that array is the same under s_1 and s_2, so $E_{s_1}[\varphi] = E_{s_2}[\varphi]$. Under s_2 the deviation $\Delta_c(g,t)$ is identically zero for $t < g$, so s_2 lies in the null of any test whose null hypothesis restricts pre-treatment deviations. If φ has size α over that null then $E_{s_2}[\varphi] \leq \alpha$, with equality when the test is similar. Hence $E_{s_1}[\varphi] \leq \alpha$, and equal to α for a similar test, although s_1 carries a recording response of arbitrary magnitude. The same argument applies to placebo in time statistics and to leads and lags coefficient vectors, which are also functions of the observed panel and the cohort labels. □

Proof of Proposition 1. Adopt the normalization $\delta_{-1} = 0$ and let the pre-period first differences be $\delta_{s+1} - \delta_s$ for $s \le -2$, which are the differences a relative magnitudes calibration reads. By Lemma 1 the deviation equivalent to the recording response is $\delta_\ell = m_c(\ell)\mathbf{1}\{\ell \ge 0\}$, so $\delta_\ell = 0$ for every $\ell \le -1$ and every pre-period first difference is exactly zero. Hence $\max_{s \le -2}|\delta_{s+1} - \delta_s| = 0$, and the defining constraint of (3.3) reads $|\delta_{\ell+1} - \delta_\ell| \le \bar{M} \cdot 0 = 0$ for every $\ell \ge -1$ and every $\bar{M} \ge 0$. Together with $\delta_{-1} = 0$ this forces $\delta_\ell = 0$ for all ℓ, so $\Delta^{\mathrm{RM}}(\bar{M}) = \{0\}$ irrespective of $\bar{M}$.

The identified set for the target under a deviation class Δ is $\{\, l'\beta_{\mathrm{post}} - l'\delta_{\mathrm{post}} : \delta \in \Delta \,\}$ with δ_{pre} pinned at its population value. With $\Delta^{\mathrm{RM}}(\bar{M}) = \{0\}$ this is the singleton $\{\, l'\beta_{\mathrm{post}} \,\}$. By Lemma 1, $\beta_c(\ell) = \vartheta_c(\ell) + m_c(\ell)$, so the singleton is $\{\vartheta_c(\ell) + m_c(\ell)\}$ and excludes $\vartheta_c(\ell)$ whenever $m_c(\ell) \ne 0$. Since the set is a point, no value of $\bar{M}$ returns an interval containing the latent effect, and no breakdown value exists. □

Proof of Lemma 2. Under Assumption 2 the regressor block in (4.1) is common across categories after the two-way within transformation, so stacking the C categories gives design matrix $[\, \mathbf{1}_C \otimes \tilde{X},\ \delta \otimes \tilde{X} \,]$ with $\delta = (\delta_1, \ldots, \delta_C)'$. Because $\mathbf{1}_C'\delta = 0$ by the centering of δ, the two blocks are orthogonal, and the normal equations separate into $(\mathbf{1}_C'\mathbf{1}_C)S\hat{\beta} = \sum_c \tilde{X}'\tilde{y}_c$ and $(\delta'\delta)S\hat{\gamma} = \sum_c \delta_c \tilde{X}'\tilde{y}_c$. Dividing by C and $\sum_c \delta_c^2$ respectively and using $b_c = S^{-1}\tilde{X}'\tilde{y}_c$ gives the stated expressions. □

Proof of Proposition 2. By (5.1) the CR1 covariance of $\hat{\gamma}$ on the post block is $\hat{V}_{PP} = \omega \sum_g \eta_{gP}^f \eta_{gP}^{f\prime}$ with $\omega = \omega_2^{-1} G/(G-1)$. The within category residual is $e_{gc} = u_{gc} - \tilde{X}_g S^{-1}\tilde{X}'u_c$ by Lemma 2, so $a_g^f = a_g - S_g S^{-1}\sum_h a_h$ and $\eta_g^f = \eta_g - S^{-1}S_g \sum_h \eta_h$. Under Assumption 3(ii), $S^{-1}S_g = I_q/G$, so $\eta_g^f = \eta_g - \bar{\eta}$, the restriction to P commutes with the correction, and $\sum_g \eta_{gP}^f \eta_{gP}^{f\prime} = (G-1)\hat{S}_P$ while $\hat{\gamma}_P - \gamma_P = G\bar{\eta}_P$. Substituting,

$$W = \hat{\gamma}_P' \hat{V}_{PP}^{-1} \hat{\gamma}_P = \omega_2\, \bar{\eta}_P' \Big[\tfrac{1}{G-1} \textstyle\sum_g (\eta_{gP} - \bar{\eta}_P)(\eta_{gP} - \bar{\eta}_P)' \Big]^{-1} \bar{\eta}_P\, G = \omega_2 T^2,$$

where the centering is the within category fitting correction and not a consequence of imposing (4.2). This is an algebraic identity and does not use Assumption 3(i).

For the distribution, Assumption 3(i) gives

$$\sqrt{G}\,\bar{\eta}_P \sim N(0, \Sigma_{PP}), \qquad (G-1)\hat{S}_P \sim W_m(\Sigma_{PP}, G-1),$$

independently, by Cochran's theorem applied to the i.i.d. Gaussian sample $\eta_{1P}, \ldots, \eta_{GP}$. Hence T^2 is Hotelling's T^2 with dimension m and $G-1$ degrees of freedom (Hotelling, 1931), and the standard transformation $\frac{G-m}{(G-1)m} T^2 \sim F(m, G-m)$ holds for $G - m \ge 1$. Substituting $T^2 = W/\omega_2$ gives (5.2).

For the last claim, write $\sum_g \eta_{gP}^f \eta_{gP}^{f\prime} = E'E$ with E the $G \times m$ matrix whose gth row is $\eta_{gP}^{f\prime}$. Summing $\eta_g^f = \eta_g - S^{-1}S_g \sum_h \eta_h$ over g and using $\sum_g S_g = S$ gives $\sum_g \eta_g^f = 0$, so the columns of E are orthogonal to $\mathbf{1}_G$ and $\operatorname{rank} E \le \min(m, G-1)$. If $G - 1 < m$ the meat is singular, $\hat{S}_P$ is not invertible, and the Hotelling representation is unavailable. □

Proof of Corollary 2. By (5.2) the exact size of the nominal chi-square test is

$$\Pr\{F(m, G-m) > (G-m)c_m/(\omega_2(G-1)m)\}, \qquad c_m = \chi^2_{m,1-\alpha}.$$

Write $F(m, G-m) \overset{d}{=} (\chi^2_m/m)/(\chi^2_{G-m}/(G-m))$ with the two chi squares independent, so the size equals $\Pr\{\chi^2_m > c_m\, \omega_2^{-1}(G-1)^{-1}\chi^2_{G-m}\}$.

For (i), write $W \overset{d}{=} R\,\chi^2_m$ with $R = \omega_2(G-1)/\chi^2_{G-m}$ independent of χ^2_m, so the size is $\mathbb{E}[h(R)]$ with $h(x) = \Pr\{\chi^2_m > c_m/x\}$ and $h(1) = \alpha$. Put $\nu = G-1$ and $d = G-m$. At fixed m, $\mathbb{E}[R-1] = (m+1)/\nu - (1-\omega_2) + O(\nu^{-2})$ and $\mathbb{E}[(R-1)^2] = 2/\nu + O(\nu^{-2})$, using $\mathbb{E}[\chi_d^{-2k}] = \prod_{j=1}^{k}(d-2j)^{-1}$ for $d > 2k$ and $G(1-\omega_2) = O(1)$. The same formula gives the third central moment of $1/\chi^2_d$ as $16/\{(d-2)^3(d-4)(d-6)\}$ for $d > 6$, whence the signed third moment $\mathbb{E}[(R-1)^3] = O(\nu^{-2})$.

A second order expansion does not deliver the remainder. Bounding it by $\sup|h'''|\,\mathbb{E}|R-1|^3$ gives $O(\nu^{-3/2})$, because the absolute third moment, unlike the signed one, gains nothing from cancellation. Expand instead to third order with a fourth order Lagrange remainder. Outside $|R-1| \le 1/2$ the contribution to $\mathbb{E}[h(R)]$ is exponentially small in ν, since h is bounded by one and the chi square Chernoff bound gives $\Pr(|R-1| > 1/2) \le e^{-c\nu}$ for some $c > 0$. Inside, $\sup_{|x-1|\le 1/2}|h''''(x)|$ is finite and $\mathbb{E}[(R-1)^4] = 3\{\mathbb{E}[(R-1)^2]\}^2 + O(\nu^{-3}) = 12/\nu^2 + O(\nu^{-3})$, so the third order term and the remainder are both $O(\nu^{-2})$.

Substituting the moments with $h'(1) = c_m f_m(c_m)$ and $h''(1) = -c_m^2 f'_m(c_m) - 2c_m f_m(c_m)$, and using $f'_m(x) = f_m(x)\{(m/2-1)/x - 1/2\}$ so that $c_m^2 f'_m(c_m) = c_m f_m(c_m)\{(m/2-1) - c_m/2\}$, the bracket collapses to

$$h'(1)\,\mathbb{E}[R-1] + \tfrac{1}{2}h''(1)\,\mathbb{E}[(R-1)^2] = c_m f_m(c_m)\Big[\frac{m+c_m}{2\nu} - (1-\omega_2)\Big],$$

which is the stated expansion. At the design of Section 9 with $G = 51$ and $C = 3$, where $\omega_2 = 0.9981$, the two terms in that bracket are 0.354 and 1.9×10^{-3} at $m = 13$ and $\alpha = 0.05$, and $G(1-\omega_2)$ is at most 0.098 over the twelve designs of Table 2 (`results/inference/anatomy_table.csv`).

For (ii), take logarithms. The event is $\log\chi^2_m - \log\chi^2_{G-m} > \log c_m - \log(\omega_2(G-1))$. With $m \to \infty$, $\log\chi^2_m$ is asymptotically normal with mean $\log m - 1/m + O(m^{-2})$ and variance $2/m$, and similarly for $\log\chi^2_{G-m}$ with $G-m \to \infty$. Standardizing and using $\log c_m = \log m + \sqrt{2/m}\,z_{1-\alpha} + o(m^{-1/2})$ gives size $\to \Phi(\chi - z_{1-\alpha})$ with $\chi = \lim\sqrt{m/2}\,\log\{(G-1)/(G-m)\}$. Expanding $\log\{(G-1)/(G-m)\} = (m-1)/(G-1) + O((m/G)^2)$ gives $\chi_{m,G} = m^{3/2}/(\sqrt{2}G) + o(1)$ when $m^{5/2}/G^2 \to 0$. If $m/G \to r \in (0,1)$ then $\log\{(G-1)/(G-m)\} \to -\log(1-r) > 0$ and $\chi_{m,G} \to \infty$, so the size tends to one. □

Proof of Lemma 3. Under Assumption 2 the two-way within transformation is the same operator for every category, so relabelling the arrays gives $\hat{\Sigma}(\mathcal{D}_\sigma)_{ck} = \hat{\Sigma}_{\sigma(c)\sigma(k)}$ and hence $L(\mathcal{D}_\sigma)_{ck} = L_{\sigma(c)\sigma(k)}$. Substituting $j = \sigma(k)$ maps $\{k \ne c\}$ bijectively onto $\{j \ne \sigma(c)\}$, and substituting $(j,l) = (\sigma(k), \sigma(l))$ maps $\{k \ne l,\ k, l \ne c\}$ bijectively onto $\{j \ne l,\ j, l \ne \sigma(c)\}$, so both averages in (6.1) are evaluated at the relabeled index and $\hat{a}_c(\mathcal{D}_\sigma) = \hat{a}_{\sigma(c)}(\mathcal{D})$. The geometric mean scaling divides by $\exp\{C^{-1}\sum_j \hat{a}_j\}$, which is σ-invariant, so $\hat{\rho}_c(\mathcal{D}_\sigma) = \hat{\rho}_{\sigma(c)}(\mathcal{D})$. The retained set and the normalized arrays are therefore relabeled with the loadings. □

Proof of Theorem 1. The argument is the classical group invariance construction; see Lehmann and Romano (2005, Ch. 15). Order invariance of $\mathcal{T}$ gives, for every $\pi \in S_C$, $\mathcal{T}(\mathcal{D}, \delta_\pi) = \mathcal{T}(\mathcal{D}_{\pi^{-1}}, \delta)$ where $\mathcal{D}_{\pi^{-1}} = (\mathcal{Y}_{\pi^{-1}(1)}, \ldots, \mathcal{Y}_{\pi^{-1}(C)})$: the statistic is computed from the multiset $\{(\delta_{\pi(c)}, \mathcal{Y}_c)\}_c$, and reindexing by $j = \pi(c)$ turns this into $\{(\delta_j, \mathcal{Y}_{\pi^{-1}(j)})\}_j$.

Under Assumption 4, $\mathcal{D}_{\pi^{-1}} \overset{d}{=} \mathcal{D}$ conditional on $\mathcal{Z}$ for each fixed π. The invariance the argument needs is under right translation of the index. Writing $f(\cdot) = \mathcal{T}(\cdot, \delta)$, so that $T(\pi) = f(\mathcal{D}_{\pi^{-1}})$, the substitution $\pi \mapsto \pi\sigma$ gives

$$T(\pi\sigma) = f\big(\mathcal{D}_{(\pi\sigma)^{-1}}\big) = f\big(\mathcal{D}_{\sigma^{-1}\pi^{-1}}\big) = f\big((\mathcal{D}_{\sigma^{-1}})_{\pi^{-1}}\big),$$

so $(T(\pi\sigma))_\pi$ is the same fixed measurable function of $\mathcal{D}_{\sigma^{-1}}$ that $(T(\pi))_\pi$ is of $\mathcal{D}$. By Assumption 4, $\mathcal{D}_{\sigma^{-1}} \overset{d}{=} \mathcal{D}$ conditional on $\mathcal{Z}$, whence $(T(\pi\sigma))_\pi \overset{d}{=} (T(\pi))_\pi$ for every $\sigma \in S_C$. Left translation does not have this property, since $T(\sigma\pi) = f(\mathcal{D}_{\pi^{-1}\sigma^{-1}})$ is not a fixed function of any one relabelling of $\mathcal{D}$, and it is right translation that the counting below uses.

We bound $\Pr(\hat{p} \le \alpha)$ for the $\hat{p}$ of Definition 2 directly, which covers ties. For $\sigma \in S_C$ put $N(\sigma) = \#\{\pi' \in S_C : T(\pi') \ge T(\sigma)\}$ and $\mathcal{A} = \{\sigma : N(\sigma) \le \alpha M\}$, so the test rejects exactly on the event $\mathrm{id} \in \mathcal{A}$. If $\mathcal{A}$ is not empty, let σ^* attain the smallest value of T on it. Every $\sigma \in \mathcal{A}$ has $T(\sigma) \ge T(\sigma^*)$, so $\mathcal{A} \subseteq \{\pi' : T(\pi') \ge T(\sigma^*)\}$ and $|\mathcal{A}| \le N(\sigma^*) \le \alpha M$, hence $|\mathcal{A}| \le \lfloor \alpha M \rfloor$ because $|\mathcal{A}|$ is an integer. Write $\phi(t) = \mathbf{1}\{\#\{\pi : t_\pi \ge t_{\mathrm{id}}\} \le \alpha M\}$ for a vector t indexed by S_C, so that the rejection indicator is $\phi\big((T(\pi))_\pi\big)$. Because $\pi \mapsto \pi\sigma$ is a bijection of S_C, the vector $(T(\pi\sigma))_\pi$ carries $T(\sigma)$ in its identity coordinate and the same multiset of values, so $\phi\big((T(\pi\sigma))_\pi\big) = \mathbf{1}\{N(\sigma) \le \alpha M\}$, and summing over σ gives $\sum_{\sigma \in S_C} \phi\big((T(\pi\sigma))_\pi\big) = |\mathcal{A}|$. Taking expectations conditional on $\mathcal{Z}$ and applying exchangeability to each term,

$$M \Pr(\hat{p} \le \alpha \mid \mathcal{Z}) = \mathbb{E}\big[\, |\mathcal{A}| \,\big|\, \mathcal{Z}\big] \le \lfloor \alpha M \rfloor,$$

so $\Pr(\hat{p} \le \alpha \mid \mathcal{Z}) \le \lfloor \alpha M \rfloor / M \le \alpha$. Averaging over $\mathcal{Z}$ preserves the bound.

Two cases sharpen it. When the M values are almost surely distinct, N takes each of $1, \dots, M$ exactly once, so $|\mathcal{A}| = \lfloor \alpha M \rfloor$ and the bound holds with equality. When they fall into $M/\mathfrak{m}$ groups of exactly $\mathfrak{m}$ equal values, distinct across groups, which is what Proposition 3(ii) gives with $\mathfrak{m} = 2$ for the two sided Wald statistic, N takes each of $\mathfrak{m}, 2\mathfrak{m}, \dots, M$ with multiplicity $\mathfrak{m}$, so $|\mathcal{A}| = \mathfrak{m}\lfloor \alpha M/\mathfrak{m} \rfloor$ and $\Pr(\hat{p} \le \alpha) = \mathfrak{m}\lfloor \alpha M/\mathfrak{m} \rfloor / M$. In either case the randomized version rejects with probability one on $\{\hat{p} \le \alpha_-\}$, whose probability is α_-, and with probability $(\alpha - \alpha_-)/(\alpha_+ - \alpha_-)$ on $\{\hat{p} = \alpha_+\}$, whose probability is $\alpha_+ - \alpha_-$, so its size is α.

For the Monte Carlo reference set of Definition 2, replace S_C by $\mathcal{R} = \{\mathrm{id}, \pi_2, \dots, \pi_B\}$. Because $\pi_2, \dots, \pi_B$ are i.i.d. uniform on S_C and drawn independently of the data, the vector $(T(\pi))_{\pi \in \mathcal{R}}$ is exchangeable conditional on $\mathcal{Z}$ under Assumption 4, and the rank argument above gives $\Pr(\text{reject}) \le \alpha$ with equality at $\mathfrak{m}\lfloor \alpha B/\mathfrak{m} \rfloor / B$. □

Proof of Proposition 3. (i) Every $\pi \in \mathrm{Stab}(\delta)$ satisfies $\delta_\pi = \delta$ and hence $T(\pi) = T(\mathrm{id})$, so the numerator of $\hat{p}$ is at least $|\mathrm{Stab}(\delta)|$ on the exhaustive reference set, and $\mathrm{Stab}(\delta)$ is the direct product of the symmetric groups on the tied groups of d, of order the product of their factorials. On the Monte Carlo reference set the identity alone gives $\hat{p} \ge 1/B$.

(ii) Because the multiset $\{\delta_c\}$ is symmetric about zero, pick an involution $r \in S_C$ matching each c with $\delta_c = v > 0$ to some c' with $\delta_{c'} = -v$ and fixing every c with $\delta_c = 0$; then $\delta_{r(c)} = -\delta_c$ for all c, and $r \ne \mathrm{id}$ because δ is not identically zero. With equally spaced d we may write $\delta_c = s\,(c - \frac{C+1}{2})$ for a scale $s > 0$ and take $r(c) = C + 1 - c$. Precomposition negates the label vector: $(\delta_{r\circ\pi})_c = \delta_{r(\pi(c))} = -\delta_{\pi(c)}$, so $\delta_{r\circ\pi} = -\delta_\pi$. Evenness of $\mathcal{T}$ then gives $T(r \circ \pi) = T(\pi)$ for every π. The map $\pi \mapsto r \circ \pi$ is left translation by an involution other

than the identity, hence a fixed point free involution on S_C, so the M values fall into $M/2$ equal pairs, and $r \circ \mathrm{Stab}(\delta)$ is disjoint from $\mathrm{Stab}(\delta)$ because $r \notin \mathrm{Stab}(\delta)$. The numerator of $\hat{p}$ is therefore at least $2|\mathrm{Stab}(\delta)|$.

For the Wald statistic, $\hat{\gamma}(-\delta) = -\hat{\gamma}(\delta)$ by Lemma 2, each cluster score is odd in δ, and the CR1 covariance is quadratic in the cluster scores, so $\hat{V}_\gamma(-\delta) = \hat{V}_\gamma(\delta)$ and the quadratic form is unchanged.

(iii) Rejection at level α requires some attainable $\hat{p} \leq \alpha$, hence $2|\mathrm{Stab}(\delta)|/C! \leq \alpha$ in the even case and $|\mathrm{Stab}(\delta)|/C! \leq \alpha$ otherwise, which reduce to $2/C! \leq \alpha$ and $1/C! \leq \alpha$ when d has no ties. □

Proof of Theorem 2. For the first claim, replace S_C in the proof of Theorem 1 by the finite group $\mathcal{E} = \{-1, +1\}^G$ acting by coordinatewise sign change. That proof uses only finiteness of the group and invariance of the data law under right translation by each of its elements, both of which the sign symmetry hypothesis supplies. $\mathcal{E}$ is abelian, so the distinction between left and right translation that the proof of Theorem 1 turns on is vacuous here.

For the second claim, let u_{gc} denote the cluster-g category-c errors under (4.2), taken before the two-way within transformation, and build the infeasible contribution from them,

$$\zeta_g^0 = \frac{\left(S^{-1} \sum_c \delta_c \tilde{X}_g' u_{gc}\right)_P}{\sum_c \delta_c^2}.$$

Using the raw rather than the within transformed errors matters here and costs nothing. It matters because the two-way within projection subtracts period means, which pool across clusters, so contributions formed from the within transformed errors are dependent across g. It costs nothing because $\tilde{X}'\Pi = \tilde{X}'$, so the aggregate is unchanged and $\sum_g \zeta_g^0 = \hat{\gamma}_P - \gamma_P$ still holds. Each ζ_g^0 is then a fixed linear functional of cluster g's errors alone, so independence across g together with symmetry of each summand about zero gives $(\epsilon_g \zeta_g^0)_g \overset{d}{=} (\zeta_g^0)_g$ for every $\epsilon \in \mathcal{E}$.

For the third claim, write z_g^r for the cluster-g score of the γ block from the fit with (4.2) imposed, so the feasible contribution is $\zeta_g = (S^{-1} z_g^r)_P / \sum_c \delta_c^2$. It replaces the errors by the restricted residuals, which subtracts a term $S^{-1} S_g [\sum_h (\cdot)_h]$. The factor $\|S_g\|/\|S\|$ is $O(G^{-1})$ and it multiplies a sum of G cluster contributions, which is $O_p(G^{1/2})$ times a single one; relative to ζ_g itself the discrepancy is therefore $O_p(G^{-1/2})$.

Asymptotic level α does not follow from an approximate symmetry argument over $\mathcal{E}$. The results of Canay et al. (2017) hold for a randomization group whose cardinality is fixed in the sample size, and $|\mathcal{E}| = 2^G$ grows with G. We argue it instead as a statement about the conditional law of the randomization distribution. Assume (W1) the cluster blocks $(\tilde{X}_g, \{u_{gc}\}_c)$ are independent across g with $\mathbb{E}[z_g^0] = 0$ under (4.2) and uniformly bounded moments of order strictly greater than four, where $z_g^0 = \sum_c \delta_c \tilde{X}_g' u_{gc}$; (W2) $\max_g \|S_g\|/\|S\| \to 0$ and $G^{-1} S \to \bar{S} \succ 0$; and (W3) $G^{-1} \sum_g \mathbb{E}[z_g^0 z_g^{0\prime}] \to \Sigma_z \succ 0$ on the block P. The sign-change distribution is the null imposed Rademacher wild cluster bootstrap of Cameron et al. (2008) in score form. Conditional on the data, $G^{-1/2} \sum_g \epsilon_g z_g^r$ is a sum of G independent mean zero terms with conditional variance $G^{-1} \sum_g z_g^r z_g^{r\prime}$, which converges in probability to Σ_z under (W1) to (W3), and (W1) with (W2) gives the conditional Lyapunov condition. The conditional bootstrap central limit theorem of Djogbenou et al. (2019) then gives the

randomization distribution of the statistic the same χ^2_m limit, conditionally in probability, that the statistic itself has under (4.2). The randomization p value is therefore asymptotically uniform. □

Proof of Corollary 3. Under the alternative $\gamma = \kappa k$, Lemma 2 gives $\hat{\gamma} = \sum_c \delta_c b_c / \Sigma^2_\delta$ with $\mathbb{E}[b_c] = (\beta_\ell)_{\ell \in L} + \delta_c \kappa k$, and with homoskedastic categories independent across c, $\mathrm{Var}(b_c) = \sigma^2 S^{-1}$. Since $\sum_c \delta_c = 0$, $\hat{\gamma}$ has mean κk and covariance $(\sigma^2/\Sigma^2_\delta) S^{-1}$, so on the post-treatment block $V = \mathrm{Var}(\hat{\gamma}_P) = (\sigma^2/\Sigma^2_\delta)[S^{-1}]_{PP}$ and, by the definition of I_k in Section 8, $\kappa^2 k_P' V^{-1} k_P = \kappa^2 \Sigma^2_\delta I_k / \sigma^2$.

The first expression in (8.2) is the minimum detectable response of the infeasible Wald statistic that uses V in place of an estimate. That statistic is noncentral chi-square with m degrees of freedom and noncentrality $\lambda = \kappa^2 \Sigma^2_\delta I_k / \sigma^2$; setting $\lambda = \lambda^*(\alpha, \beta, m)$ and solving for κ gives it. The feasible CR1 Wald statistic does not have that law. The identity in the proof of Proposition 2 uses no distributional assumption and holds under the alternative with $\bar{\eta}_P$ replaced by $\hat{\gamma}_P / G$, so under Assumptions 2 and 3 the feasible statistic is ω_2 times a noncentral Hotelling T^2 with dimension m and $G - 1$ degrees of freedom. Equation (8.2) therefore understates the minimum detectable response of the feasible test twice, through the critical value, as Section 8 records, and through the distribution under the alternative.

For the directed statistic, t is standard normal under (4.2) and normal with unit variance and mean $\kappa \sqrt{\Sigma^2_\delta I_k} / \sigma$ under the alternative, so power $1 - \beta$ at level α requires that mean to equal $z_{1-\alpha} + z_\beta$, which gives the second expression. □

Proof of Theorem 3. Two configurations $(b_c)_c$ with the same contrast vector $(b_c - \bar{b})_c$ differ by a shift common to all categories, so the $C - 1$ contrasts are a maximal invariant and an invariant test is a function of them. The maximal invariant is the full contrast vector, of dimension $(C-1)q$, and not $\hat{\gamma}_P$. Decompose the contrast space orthogonally in the category index into the direction δ and its complement. The δ-component is $\hat{\gamma} = \sum_c \delta_c b_c / \sum_c \delta_c^2$, with mean κk. The remaining $C - 2$ contrasts have mean zero for every κ, so they are ancillary for κ, and because Σ does not vary with c they are uncorrelated with $\hat{\gamma}$ and, being jointly Gaussian with it, independent of it. Write A for that block.

Conditional on $A = a$, the law of $\hat{\gamma}_P$ is $N(\kappa k_P, V)$ and does not depend on a. Its density in κ is proportional to $\exp\{\kappa k_P' V^{-1} \hat{\gamma}_P - \frac{1}{2} \kappa^2 k_P' V^{-1} k_P\}$ relative to the $\kappa = 0$ density, a one parameter exponential family with natural parameter κ and natural sufficient statistic $U = k_P' V^{-1} \hat{\gamma}_P$, which has monotone likelihood ratio in κ. By the Karlin–Rubin theorem (Lehmann and Romano, 2005) the conditional test rejecting for large U is uniformly most powerful at every conditional level, and at conditional level u its power is $\Psi(u) = \Phi(\mu + \Phi^{-1}(u))$ with $\mu = \kappa \sqrt{k_P' V^{-1} k_P} > 0$.

An invariant test may still randomize its level over the ancillary block, taking conditional level $\alpha(A)$ with $\mathbb{E}[\alpha(A)] = \alpha$, and ruling that out is the remaining step. Differentiating, $\Psi'(u) = \exp\{-\mu^2/2 - \mu \Phi^{-1}(u)\}$, which is strictly decreasing in u, so Ψ is strictly concave on $(0, 1)$ and Jensen's inequality gives $\mathbb{E}[\Psi(\alpha(A))] \leq \Psi(\alpha)$, with equality if and only if $\alpha(\cdot)$ is almost surely constant. The unconditional power of an invariant level-α test is therefore at most $\Psi(\alpha)$, attained by $\alpha(\cdot) \equiv \alpha$. Standardizing U by $\sqrt{k_P' V^{-1} k_P}$ gives $t \sim N(\kappa \sqrt{k_P' V^{-1} k_P}, 1)$, from which the critical value and the stated power function follow. □

S2. ADDITIONAL SIMULATION EVIDENCE

The exact null law in the ideal experiment. Assumption 3 is not satisfied by the panel design, so Proposition 2 is checked separately in an experiment built to satisfy it: cluster contributions drawn i.i.d. $N(0, \Sigma)$ with $m = 13$, Σ redrawn for each of three seeds, and 60,000 replications in total at each G. The identity $W = \omega_2 T^2$ holds to a maximum relative error of 4.0×10^{-14} at $G = 20$ and 1.3×10^{-15} at $G = 100$. Referring the statistic to the F law of (5.2) at the nominal G gives null rejection rates of 0.0495, 0.0489, 0.0502 and 0.0499 at $G = 20$, 30, 51 and 100, quoted to four decimals here because the claim is that they are indistinguishable from the nominal 0.05 rather than merely close to it. Kolmogorov–Smirnov tests of the resulting p values against the uniform return $p = 0.81$, 0.02, 0.06 and 0.77 at those four G, so the pooled $G = 30$ draws reject uniformity at the 5% level. The closed-form chi-square size implied by (5.2) matches the simulated one to within 0.003 at every G. Source: `anatomy_identity.csv`.

Accuracy of the substitution of G_{eff}. Evaluating (5.2) at G_{eff} from Definition 1 predicts a null rejection rate of 0.378 against the observed 0.389 at $G = 51$, $C = 5$, $m = 13$. Across the twelve (G, C) cells the mean absolute prediction error is 2.37 percentage points using G_{eff} and 13.13 percentage points using G. In this design $G_{\text{eff}} = 36.42$, or 71.4% of G, and it is a design object rather than a random one: its standard deviation across replications is 10^{-13}. The Hotelling calibration evaluated at G_{eff} tracks the nominal level only in the middle of the grid. It rejects at 0.000 in all three cells at $G = 20$, at 0.022, 0.020 and 0.021 at $G = 30$, at 0.047, 0.045 and 0.045 at $G = 51$, and at 0.061, 0.065 and 0.055 at $G = 100$, so the substitution is conservative at twenty clusters and mildly liberal at a hundred.

When exchangeability fails. A departure from Assumption 4 is priced in a Gaussian limit experiment rather than in the panel: one $N(0, \omega_c)$ draw per category, a single restriction, the same randomization over the $C!$ label assignments, a covariance estimate that treats the categories as homoskedastic, and 20,000 replications. With the inflated variance on the highest exposure category, the label permutation test using the Wald statistic rejects at 0.050, 0.064, 0.077, 0.090 and 0.102 at $C = 5$, and at 0.048, 0.062, 0.079, 0.100 and 0.125 at $C = 8$, for variance ratios 1, 2, 4, 9 and 25. Placing the inflated variance on the lowest exposure category instead moves those rates by at most 0.009. The distortion is monotone in the ratio, and its largest value in the grid stays below the chi-square rate of 0.356 measured at $G = 51$, $C = 8$, $m = 13$. DiCiccio and Romano (2017) show for a regression coefficient that studentizing the statistic makes the permutation test exact under independence and asymptotically valid under heteroskedasticity, so a studentized category level statistic is the natural repair for this departure; we have not derived its behavior under Assumption 4.

Two further experiments show that the rates above overstate what the departure costs the procedure we run. The homoskedastic covariance estimate of the limit experiment is not the one the panel uses. The CR1 meat $\sum_g \eta^f_{gP} \eta^{f'}_{gP}$ is quadratic in the realized residuals, which enter it category by category weighted by $\delta_{\pi(c)}$, so its variance estimate already carries a separate contribution per category and already moves with π. Supplying the limit experiment with a variance estimate of the same kind, $\hat{\omega}_c = \omega_c \chi^2_\nu / \nu$ drawn independently across c, reverses the sign of the distortion: with the inflated variance on the highest exposure category

the studentized Wald statistic rejects at 0.050, 0.039, 0.028, 0.015 and 0.006 at $C = 5$ and at 0.048, 0.040, 0.030, 0.018 and 0.005 at $C = 8$ as ν grows without bound, over the same grid. Studentizing leaves the exchangeable case at nominal size and makes the departure conservative rather than liberal.

The panel behaves the same way. Running the same departure through the design of this section with the CR1 statistic itself, drawing category error standard deviations that deliver median realized residual variance ratios of 2.77, 3.07, 5.09, 11.22 and 32.90 over 2,000 replications per cell, the two-sided label permutation test rejects at 0.057, 0.045, 0.032, 0.020 and 0.009 at $C = 5$ and at 0.054, 0.048, 0.038, 0.027 and 0.009 at $C = 8$, with a Monte Carlo standard error of at most 0.005 across the forty randomization cells. The directed version rejects at 0.051, 0.040, 0.032, 0.023 and 0.010 and at 0.050, 0.046, 0.041, 0.033 and 0.016. No cell exceeds the nominal level, and the rates track the studentized limit experiment at $\nu = 10$, which gives 0.051, 0.042, 0.031, 0.020 and 0.009 at $C = 5$. Against a category variance ratio of 33 the cost of the departure is power and not level. Source: `studentized_check.csv`.

In the limit experiment with a homoskedastic covariance estimate the sign of the distortion depends on how the variance spread lines up with the exposure index. In the implementation used for Tables 2 and 3, which recomputes the CR1 covariance for every permuted label vector, it does not. Moving the inflated variance to the lowest exposure category instead of the highest gives 0.057, 0.039, 0.028, 0.024 and 0.015 at $C = 5$ and 0.054, 0.043, 0.033, 0.031 and 0.011 at $C = 8$, so the panel rejection rate falls with the variance ratio under either placement. A separate $C = 6$ design with category variances decreasing in exposure moves the same way, from 0.047 at variance ratio 1 to 0.021 at ratio 25 over 2,000 replications. Sources: `exchangeability_failure.csv` and `studentized_check.csv`, and `a2_size_heteroskedastic.csv` under `results/inference/`.

The price of label randomization. Relative to a normal benchmark at level 0.05 and power 0.80, the label permutation test requires the standardized signal inflated by a factor of 1.892 at $C = 4$, 1.352 at $C = 5$, 1.235 at $C = 6$, 1.149 at $C = 8$ and 1.098 at $C = 10$, so the price falls as C grows. The factor is a large-sample calculation and it holds up only at the larger C. The minimum detectable response it implies is 0.231 at $C = 8$ against 0.243 in simulation, and 0.327 at $C = 5$ against 0.477, where the simulated value is 46% larger, at the smallest C for which Proposition 3(iii) makes the two-sided test usable. Sources: `perm_power_ceiling.csv` and `perm_mde_vs_sim.csv`.

Exposure spread, panel length and the pre window. Three further blocks vary the design rather than the procedure, at $G = 51$ and $C = 5$. Compressing the exposure indices from $[0.1, 1]$ to $[0.4, 0.6]$ cuts Σ_δ^2 from 0.506 to 0.025 and raises the closed-form minimum detectable response of the Wald test from 0.363 to 1.639, a factor of 4.5, with the directed test moving from 0.242 to 1.854. Halving the panel to $T = 60$ moves the Wald figure by under 2%, from 0.363 to 0.370. Replacing the six period pre window by the single omitted period $\ell = -1$ raises I_k from 23.6 to 448 and cuts the closed-form directed minimum detectable response to 0.050, against 0.081 for the simulated directed cluster sign-change test; the chi-square null rejection rate in that block is 0.405 over 600 replications, so narrowing the pre window buys power without repairing the calibration. Sources: `mde_table.csv` and

`reference_block_table.csv`.

Robustness. Four designs drop the features of the block above that favour the procedures we propose, each at $G = 51$, $C = 5$ and 1,000 replications, against a Monte Carlo standard error of 0.007 at a rate of 0.05. The first gives every cluster five units and a cluster by period shock with standard deviation 0.6, so that the clustering carries dependence rather than labelling singletons. The second adds unequal cluster sizes, spread geometrically from two units to sixteen at the same total, which is the configuration MacKinnon and Webb (2017) identify as the wild bootstrap's hardest case. The third replaces the independent errors by an AR(1) in the period index with coefficient 0.7. The fourth replaces the Gaussian errors by standardized t_5 draws, so that only the tails change.

The label permutation test holds its size in all four, rejecting at 0.050, 0.058, 0.059 and 0.056 against 0.046 in the baseline design. Nothing in Theorem 1 leads one to expect otherwise, since Assumption 4 restricts the category direction alone and is untouched by cluster size, serial dependence or tails, but the measurement is what the theorem is worth if the theorem is right. The chi-square calibration over rejects in all four, between 0.307 and 0.372 against 0.394 in the baseline design.

Two procedures degrade. The restricted wild cluster bootstrap moves from 0.048 to 0.066 and 0.065 in the two clustered designs. The directed cluster sign-change test moves from 0.059 to 0.094 and 0.086, which is five Monte Carlo standard errors above nominal and is the reason the recommendation in Section 1 is qualified. Its two-sided counterpart is far less affected, at 0.057 and 0.063 against 0.050. The pattern is consistent with Theorem 2: the finite-sample statement covers the infeasible statistic, the feasible one differs by $O_p(G^{-1/2})$, and within cluster dependence is what makes that gap bite. A one-sided statistic divided by a noisier estimated standard deviation is where it shows first.

A bias-reduced variance does not repair the calibration. The literature on few clusters offers a better variance estimator than the CR1 used here: the bias-reduced linearization of Bell and McCaffrey (2002), which premultiplies each cluster's residuals by $(I - H_{gg})^{-1/2}$ before forming the meat. Applying it changes almost nothing. Across the five designs the chi-square rejection rate moves from 0.394, 0.307, 0.361, 0.358 and 0.372 under CR1 to 0.390, 0.305, 0.357, 0.353 and 0.372 under CR2, every difference inside one Monte Carlo standard error, and no cluster in any design has leverage high enough for the adjustment to be undefined. In the two applications it moves the crime chi square p value from 0.025 to 0.026 and the two health values from 0.730 and 0.391 to 0.736 and 0.435. The reason is that the two corrections address different things. Bell and McCaffrey (2002) correct the downward bias of the meat matrix, which is an $O(1/n_g)$ effect and is small when each cluster contributes many periods, while the distortion this paper is about is the substitution of a chi-square for a distribution whose denominator has $G - m$ degrees of freedom, and that survives any reweighting of the residuals. The exact law of Proposition 2 is stated for CR1 and does not transfer, so we do not report a Hotelling calibration of the CR2 statistic. Source: `robust.csv`.

Coverage of honest confidence sets under a recording response

Table S1. Coverage of the latent effect by honest confidence sets under a recording response. Single cohort, latent effect 1.0, recording wedge 0.6 at the target horizon, nominal 95%. Sets computed with a reference implementation of Rambachan and Roth (2023). $\widehat{M}$ calibrates the smoothness bound from the pre-period as the reference software's exported routine does; institutional sets it from the size of the recording change. The last column is coverage of the recorded effect, which is what the interval is built to cover. Source: `results/coverage/coverage_zero_ref.json`, produced by `scripts/coverage_zero_ref.py`.

			$\Delta^{\mathrm{RM}}(\bar{M})$			$\Delta^{\mathrm{SD}}(M)$		
N	reps	$\widehat{M}$	$\bar{M}=0.5$	$\bar{M}=1$	$\bar{M}=2$	$M=\widehat{M}$	institutional	recorded
400	400	0.7165	0.680	0.955	0.998	1.000	1.000	1.000
1,600	400	0.3615	0.048	0.427	0.873	1.000	1.000	1.000
6,400	400	0.1814	0.000	0.007	0.287	1.000	1.000	1.000
25,600	200	0.0890	0.000	0.000	0.000	1.000	1.000	1.000
102,400	100	0.0450	0.000	0.000	0.000	0.210	1.000	1.000
409,600	20	0.0226	0.000	0.000	0.000	0.000	1.000	1.000

S2.1 What coverage does

Table S1 reports coverage of the latent effect in a single cohort design with a latent effect of 1.0 and a recording wedge of 0.6 at the target horizon. Confidence sets are computed with a reference implementation of Rambachan and Roth (2023) rather than by hand. For the smoothness class it computes the optimal fixed length interval of their Section 4.1. For the relative magnitudes class it solves the identified set by their Lemma 2.2 and extends it by a normal sampling term, a construction its own documentation calls conservative relative to the conditional hybrid intervals the paper recommends; a coverage failure computed this way therefore bounds the failure of the recommended interval. The row labeled $\widehat{M}$ calibrates the smoothness bound as the reference software's exported routine does, as the maximum over one-sided upper confidence bounds for the second differences of the observed pre-period coefficients. The row labeled institutional instead sets the bound from the size of the recording change, which is the alternative this section recommends.

The pattern in Table S1 is not a small-sample artifact that more data repairs. More data is what causes it. The calibration input falls at the sampling rate while the wedge does not move, so every relative magnitudes interval eventually excludes the latent effect, and larger $\bar{M}$ only delays the crossing. The institutional bound, which never consults the pre period, is unaffected.

Two of the three blocks in Table S1 bear on different questions, and only one of them is about calibration. The relative magnitudes columns fix $\bar{M}$ in advance at 0.5, 1 and 2, so no quantity in them is chosen from the data. What fails there is not a calibration rule but the class itself: by Lemma 1 the pre-period second differences that $\Delta^{\mathrm{RM}}(\bar{M})$ scales are identically zero under a recording response, so the class is $\{0\}$ for every $\bar{M}$, which is Proposition 1. The row labeled $\widehat{M}$ is the one that reads a bound off the pre period, and the nonadaptivity result

of Low (1997) and Cai and Low (2004) is already a reason to expect it to fail; we report it because it is what the exported routine computes, not because Rambachan and Roth (2023) recommend it. Their own guidance, like that of Armstrong and Kolesár (2020), is to choose the sensitivity parameter from knowledge of the setting. The institutional row is that recommendation carried out, and it is the row that holds coverage at 1.000 throughout. The table is therefore an argument for the a priori bound rather than against the framework.

The simulation design in full

Design. The simulated panel is a two-layer recording model with count outcomes. A latent factor $F_{it} = f_i + g_t + \tau D_{it} + v_{it}$ with $\tau = 0.4$ drives C category indices $Y^*_{itc} = a_{ic} + \lambda_{tc} + \theta_c F_{it} + \varepsilon_{itc}$, in which the loadings θ_c are a uniformly random permutation of C values evenly spaced in 1 ± 0.3, drawn independently of d. Recording thins counts. With $\mu_{ic} \sim N(0, 0.5^2)$ the untreated recording rate is $r_{ic} = \min\{\max(0.6\, e^{0.2\mu_{ic}}, 0.02), 1\}$; on treated cells it becomes $\tilde{r}_{itc} = \min\{r_{ic}(1 + \kappa k(\ell_{it}) d_c), 1\}$ and equals r_{ic} elsewhere; and $Y_{itc} \sim \text{Poisson}(500\, \tilde{r}_{itc}\, e^{0.25 Y^*_{itc}})$. The regression outcome is $\log\{(Y_{itc} + 0.5)/500{,}000\}$, so the recording wedge enters (4.1) as $\log(1 + \kappa k(\ell) d_c)$ rather than as $\kappa k(\ell) d_c$. The response path is flat: $k(\ell) = 1$ at every post-treatment horizon. All disturbances are Gaussian, independent across clusters and serially independent within cluster. The null (4.2) corresponds to $\kappa = 0$.

The design below puts one unit in every cluster, draws errors independent across periods and Gaussian, and makes the category loadings exchangeable by construction. Those choices are favourable to the procedures we propose, and the paragraph headed robustness at the end of this section runs the same calibrations in four designs that drop them.

We vary $G \in \{20, 30, 51, 100\}$ with one unit per cluster, so every cluster carries the same number of observations, and $C \in \{3, 5, 8\}$ with exposure indices equally spaced on $[0.1, 1]$. Every design has $T = 120$, adoption at $t = 48$, 72 and 96 by 20% of units each and 40% never treated, which gives $G_{\text{eff}}/G = 0.714$. Six pre-treatment and twelve post-treatment horizons enter (4.1), omitting $\ell = -1$, so $q = 18$ and $m = 13$ in every cell of every block. Each size cell uses 2,000 replications at nominal level 0.05.

The wild cluster bootstrap uses 499 draws and the cluster sign-change test 999, each including the identity draw, so $\hat{p} = (1 + \#\{b \geq 2 : T_b \geq T_{\text{obs}}\})/B$. The label permutation test of Definition 2 enumerates all $C!$ assignments at $C = 3$ and $C = 5$. At $C = 8$ it is the Monte Carlo version with $B = 999$, namely 998 sampled assignments together with the identity; the largest level it can attain without exceeding $\alpha = 0.05$ is then $\lfloor \alpha B \rfloor / B = 0.049$, and its resolution floor is $1/B$ rather than the $2/C!$ of Proposition 3. The chi bar square statistic uses $B = 199$ at $C = 8$. Monte Carlo standard errors are 0.005 at a rejection rate of 0.05, 0.011 at 0.39 and 0.006 at 0.92. Output files named in this section sit in `results/inference/` unless another directory is given.

Two features of the design bear on what the tables can show. The loadings are drawn independently of d and the ε_{itc} are i.i.d. across categories, so Assumption 4 holds by construction and the permutation rows of Table 1 of the paper measure the size of a test to which Theorem 1 applies, for the null (4.2) under the randomization of Definition 2; the departure from Assumption 4 is priced separately below. Assumption 3 does not hold here: part (i) fails because the cluster scores come from Poisson counts through a nonlinear wedge, and

part (ii) fails through the never-treated share. Proposition 2 is therefore an approximation in this design, and the accuracy paragraph below reports the error. Cluster sizes are equal and no disturbance is serially correlated, so nothing in this section tests the cluster size heterogeneity that MacKinnon and Webb (2017) identify as the bootstrap's failure mode, and nothing tests within cluster serial dependence.

Table S2. Size adjusted power against a flat recording response of magnitude κ, $G = 51$

		Recording response κ							
Statistic and calibration	C	0.05	0.10	0.15	0.20	0.30	0.45	0.60	0.80
Wald, oracle critical value	5	0.082	0.107	0.145	0.212	0.362	0.705	0.828	0.872
	8	0.058	0.113	0.150	0.277	0.537	0.850	0.930	0.953
Wald, wild cluster bootstrap	5	0.080	0.110	0.147	0.220	0.362	0.715	0.830	0.872
	8	0.053	0.112	0.157	0.273	0.532	0.852	0.925	0.950
Wald, label permutation	5	0.083	0.103	0.138	0.170	0.212	0.383	0.452	0.383
	8	0.055	0.097	0.142	0.187	0.395	0.655	0.733	0.722
Directed, label permutation	5	0.147	0.250	0.403	0.518	0.657	0.797	0.815	0.772
	8	0.175	0.370	0.535	0.732	0.890	0.973	0.978	0.970
Directed, cluster sign change	5	0.143	0.320	0.527	0.688	0.895	0.983	0.987	1.000
	8	0.155	0.403	0.613	0.842	0.968	0.997	0.997	1.000

Notes: size-adjusted rejection rates against the alternative $\gamma_\ell = \kappa k(\ell)$ with $k(\ell) = 1$ at every post-treatment horizon, for the joint null $H_0 : \gamma_P = 0$ of (4.2). Each entry uses the critical value set at the empirical 5% quantile of that procedure's p value distribution under $\kappa = 0$ in the same design. Because the randomization p values have atoms, that rule does not deliver a null rate of 0.05 in every row. At $C = 5$ the achieved rates are 0.050 for the oracle Wald and the directed sign-change rows, 0.051 for the bootstrap, 0.056 for the directed permutation row and 0.060 for the two-sided permutation row, so the two permutation rows at $C = 5$ run at a higher level than the rows beside them; at $C = 8$ every row is between 0.050 and 0.051. The first row is not a feasible test: the chi-square calibration has null rejection rate 0.389 at $C = 5$, and its size-adjusted cut is a nominal p value of 7.3×10^{-5}, which cannot be computed without the null simulation. It is reported as a power benchmark. For the other rows the adjustment moves the cut only to between 0.043 and 0.067 in p value units. The directed rows use the contrast $w = \mathbf{1}_m$, which is the true response path in this block. The block also ran $C = 3$, where the only attainable two-sided permutation p values are 1/3, 2/3 and 1 and the adjusted rule reduces to rejecting at $\hat{p} = 1/3$, with an achieved null rate of 0.340; those cells are omitted. Design: $G = 51$ clusters, $m = 13$ post-treatment restrictions, $q = 18$ horizons, $T = 120$, adoption at $t = 48$, 72 and 96, exposure indices equally spaced on $[0.1, 1]$. Label permutation enumerates all $C!$ assignments at $C = 5$ and uses 998 sampled assignments together with the identity at $C = 8$; the cluster sign-change test uses 999 draws including the identity. 600 replications per cell; Monte Carlo standard error at a rate of 0.5 is 0.020. Sources: `results/inference/power_curves.csv` and `size_adjustment_audit.csv`.

Power over the full grid, and the nonmonotone cells

Power. Table S2 reports size-adjusted power at $G = 51$ against a recording response with a flat path, $k(\ell) = 1$ at every post-treatment horizon, indexed by the magnitude κ, over 600

replications per cell. Two patterns run in opposite directions. Label randomization with the two sided Wald statistic is the least powerful procedure at every κ at or above 0.10 in both C of Table 2: at $C = 5$ and $\kappa = 0.45$ it attains 0.383 against 0.705 for the Wald statistic at its oracle critical value. At $\kappa = 0.05$ the three two-sided procedures sit within one Monte Carlo standard error of one another, at 0.083, 0.082 and 0.080 at $C = 5$, so the ordering is not measured there. Directing the statistic more than reverses the loss. The directed cluster sign-change test attains 0.983 at $C = 5$ and $\kappa = 0.45$, and at $C = 8$ it reaches 0.842 at $\kappa = 0.20$ where the two sided Wald reaches 0.277. The gain comes from the direction restriction rather than from the randomization.

The directed statistics in Table S2 use the contrast $w = \mathbf{1}_m$, which is the true response path in every cell of the block, since $k(\ell) = 1$ throughout. Theorem 3 is stated for known k_P, and the table measures the directed tests under that condition. No cell was run in which the assumed direction differs from the true one, so these entries bound what a researcher who has to guess the shape can expect, and the table carries no evidence on how much of the gain survives a wrong guess.

The size-adjusted power of the two-sided permutation test is nonmonotone in κ at $C = 5$, falling from 0.452 at $\kappa = 0.60$ to 0.383 at $\kappa = 0.80$. The two rates come from independently seeded replications, so the standard error of the difference is 0.028 and the decline of 0.068 is 2.4 of them.

The same disagreement appears twice more at $C = 5$. The large-sample local power calculation for the label permutation test predicts 0.538, 0.768 and 0.927 at $\kappa = 0.45$, 0.60 and 0.80 against simulated 0.383, 0.452 and 0.383, overstatements of 15, 32 and 54 percentage points, and it predicts 0.041 at $\kappa = 0$ against the nominal level of 0.05. The minimum detectable response it implies is 0.327 against a simulated 0.477, while at $C = 8$ the same calculation gives 0.231 against 0.243. The nonmonotonicity and the theory simulation gap are the same disagreement at small C, and we have not isolated its cause. Sources: `local_power_check.csv` and `perm_mde_vs_sim.csv`.

The two sensitivity classes developed

S2.2 Sensitivity analysis calibrated from pre trends

The standard response to Corollary 1 is to abandon testing and report a confidence set that is honest over a class of deviations (Rambachan and Roth, 2023). Two classes are in common use. The smoothness class $\Delta^{\mathrm{SD}}(M)$ bounds second differences of the deviation path by M. The relative magnitudes class bounds post-treatment first differences by a multiple of the largest pre-treatment first difference,

$$\Delta^{\mathrm{RM}}(\bar{M}) = \Big\{\delta : |\delta_{\ell+1} - \delta_\ell| \le \bar{M} \max_{s \le -2} |\delta_{s+1} - \delta_s| \ \ \forall \ell \ge -1\Big\}, \tag{S2.1}$$

with the reference period normalized to $\delta_{-1} = 0$. Two index ranges carry the argument and are worth stating explicitly. The maximum on the right runs over $s \le -2$, so it uses only differences computable from pre-treatment coefficients. The constraint on the left starts at $\ell = -1$, so the first difference it binds is $\delta_0 - \delta_{-1} = \delta_0$, the step out of the reference period into

the first post-treatment period. Both conventions are those of the reference implementation, which bounds $|\delta_0|$ by the same quantity that bounds every later difference.

Substituting (3.2) into (S2.1) gives the difficulty at once. The equivalent deviation is zero throughout the pre period, so the quantity the class scales is zero, and $\Delta^{\mathrm{RM}}(\bar{M}) = \{0\}$ for every $\bar{M} \geq 0$. The family is not a nested sequence of relaxations indexed by $\bar{M}$; it is the point null repeated. There is no breakdown value, because no member of the family admits any deviation at all.

Proposition 1 is a population statement and the finite-sample version is what a researcher meets. The honest set conditions on the estimated pre-period coefficients rather than on the true deviation, so the estimated bound is $\bar{M} \max_s |\widehat{\beta}_{s+1} - \widehat{\beta}_s|$, which is positive in every realized sample. Under a recording response that bound is pure sampling noise of order $n^{-1/2}$ while the wedge it is meant to cover is of order one. The interval therefore shrinks around the contaminated estimand at the sampling rate, and coverage of the latent effect falls to zero.

Arora and Chand (2026) note the mechanism in prose, cautioning that when pre trends are uninformative Δ^{RM} produces "intervals whose width tends to zero even as the bias does not." Their remark is a paragraph in a paper about a different estimand, carries no formal statement, and treats flat pre trends as a circumstance that may or may not obtain. Assumption 1 identifies a mechanism under which it obtains exactly, always, and undetectably.

Implementation, step by step

Implementation

The three procedures are run as follows. Steps 1 and 2 use only the design and can be carried out before any outcome is observed.

1. Form $\tilde{X} = \Pi X$ and $S = \tilde{X}'\tilde{X}$ on the common cell set of Assumption 2, fix the post-treatment block P, and record $m = |P|$. For a contrast $w \in \mathbb{R}^m$ set $a = S^{-1}[\cdot, P]w$ and $\varphi_g = a' S_g a$, and evaluate $G_{\mathrm{eff}}(w)$ from Definition 1. Where one number is wanted for a joint test, take $\min_w G_{\mathrm{eff}}(w)$ over the m coordinate contrasts $w = e_1, \ldots, e_m$, which is the least favorable of the directions the joint statistic combines.

2. Evaluate (5.2) at m and at $G_{\mathrm{eff}}(w)$ in place of G to obtain the size the nominal level-α chi-square test would have. If that size is larger than the researcher is willing to carry, either reduce m until it is not, or calibrate by Steps 3 and 4.

3. With $C \geq 5$ categories, use the label permutation test of Definition 2, enumerating all $C!$ assignments while that is feasible and sampling from S_C otherwise. Proposition 3(iii) rules the two-sided version out at $\alpha = 0.05$ when $C \leq 4$, and Assumption 4 is what buys its finite-sample validity.

4. With $C \leq 4$, or when Assumption 4 is doubtful, use the cluster sign-change test of Theorem 2 or the restricted wild cluster bootstrap, both with the null (4.2) imposed on the residuals and $B - 1$ draws.

5. When the response shape k_P is known up to scale, replace the two sided Wald statistic by the directed statistic of Theorem 3 inside whichever randomization Step 3 or Step 4 selected. Table S2 reports what that substitution is worth at $G = 51$.

6. To use (5.2) as a reference distribution rather than as a diagnostic, match ω_2 to the finite-sample factor the software applies. Proposition 2 sets $\omega_2 = (n - 2q)/(n - 1)$, counting the $2q$ interacted regressors and not the absorbed unit and period effects, and the factor multiplying the sandwich is then $\{G/(G-1)\}\omega_2^{-1}$. A CR1 implementation that counts the absorbed effects reports a larger factor, so recover ω_2 by dividing $G/(G-1)$ by the factor the software used. In the test design used to check Proposition 2 the two conventions differ by a factor of 1.178 (`verification.json`, entry `V1b`).

The code implementing Steps 1 to 6 is `src/did_measurement/inference.py` in the replication archive.

The fixed contrast and the efficient statistic compared Which statistic that statement describes matters, and the two behave differently enough that the distinction is worth measuring. The cosine above governs the statistic of Theorem 3, which weights by V^{-1}. The statistic reported in Tables 1 and 2 is the fixed contrast $w'\hat{\gamma}_P/\sqrt{w'Vw}$, whose noncentrality is $\kappa w'k_P/\sqrt{w'Vw}$, so its loss relative to the optimum is $w'k_P/\{\sqrt{w'Vw}\sqrt{k_P'V^{-1}k_P}\}$, at most one by Cauchy Schwarz with equality exactly when $w \propto V^{-1}k_P$. The two ratios differ sharply. In the design of Section 9 at $G = 51$ and $C = 5$ against a flat truth, running the fixed contrast at a response growing with horizon, at one that fades, and at one confined to the second half of the post window gives ratios of 0.80, 0.78 and 0.79 against 0.80 at the truth, while the same shapes give the V^{-1} cosines of Theorem 3 as 0.46, 0.12 and 0.36 against 1.

The simulated consequences follow the ratios. Over 600 replications per cell, every one of the forty eight cells is at nominal size, between 0.057 and 0.065 for the fixed contrast and between 0.035 and 0.070 for the V^{-1} weighted statistic, against a Monte Carlo standard error of 0.009 at a rate of 0.05. That is the level statement of the previous paragraph measured rather than assumed. At $C = 5$ and $\kappa = 0.30$ the fixed contrast attains size-adjusted power 0.607 at the true shape and 0.530, 0.602 and 0.527 at the three wrong ones, a worst case loss of 0.080, while the V^{-1} weighted statistic attains 0.467 at the truth and 0.195, 0.057 and 0.212, a worst case loss of 0.410. The optimal statistic is the one a wrong shape destroys.

Two things follow. The power advantage reported in Table S2 does not depend on knowing the response path, since the statistic carrying it loses at most 0.08 of power across shapes that span the plausible range. And Theorem 3 is an optimality statement about a limit experiment with V known rather than a recommendation: with V estimated from 51 clusters on 13 restrictions the fixed contrast beats the V^{-1} weighted statistic even at the true shape, 0.607 against 0.467, because inverting an estimated V of that dimension costs more than the weighting gains. We therefore report the fixed contrast throughout and read Theorem 3 as the bound that identifies what a known V would buy. Source: `misdirection.csv`.

When k_P is restricted only to the nonnegative orthant no uniformly most powerful invariant test exists, and the likelihood ratio statistic has the chi bar square null distribution of

Kudô (1963) and Shapiro (1988), a mixture $\sum_{j=0}^{m} w_j(V)\chi^2_j$ in which $w_j(V)$ is the probability that the projection of $N(0, V)$ onto the nonnegative orthant in the V^{-1} metric has exactly j positive coordinates. In the designs of Section 9, its size-adjusted power under the label permutation calibration lies between that of the two sided Wald and that of the directed statistic at every κ on the grid and at both $C = 5$ and $C = 8$. We have not established that ordering in general.

The ratio of minimum detectable responses depends on how each test is calibrated. Taking the Wald at the exact critical value from (5.2) and the directed test at the $t_{G_{\text{eff}}-1}$ critical value that its estimated CR1 variance requires, that ratio at power 0.80 is 0.393 at $G_{\text{eff}} = 36.42$, 0.448 at $G_{\text{eff}} = 51$ and 0.516 at $G_{\text{eff}} = 100$ (`results/inference/verification.json`, entry `E11`). Calibrating the directed test at $z_{1-\alpha}$ instead, as Theorem 3 does under known V, gives 0.386, 0.442 and 0.512. At the first of these designs the direction restriction divides the $\Sigma^2_\delta I_k$ required to detect a given κ, and so the number of clusters required at fixed per cluster information, by 6.5.

The effective cluster count derived Under errors that are homoskedastic and independent across clusters, $\varphi_g / \sum_h \varphi_h$ is the share of $\text{Var}(w'\hat{\gamma}_P)$ carried by cluster g. Write $\bar{\varphi} = G^{-1}\sum_g \varphi_g$ and $\Gamma = G^{-1}\sum_g \{(\varphi_g - \bar{\varphi})/\bar{\varphi}\}^2$. Then $G/(1+\Gamma) = G\bar{\varphi}^2/(G^{-1}\sum_g \varphi_g^2) = (\sum_g \varphi_g)^2 / \sum_g \varphi_g^2$, so Definition 1 is the count $G/(1+\Gamma)$ of Carter et al. (2017) evaluated at the design weights φ_g. That paper defines Γ for the t-statistic on a scalar contrast through cluster level contributions to the variance of the estimator; Definition 1 fixes those contributions at their homoskedastic design values, which is what makes it computable before any outcome is observed. It satisfies $1 \le G_{\text{eff}}(w) \le G$, with equality on the right when the φ_g are equal, which is what Assumption 3(ii) imposes. Substituting G_{eff} for G in (5.2) is an approximation; Proposition 2 does not license it. Section 9 reports its accuracy.

Every G_{eff} quoted below is the equal weight contrast $w = \mathbf{1}_m$, and the design rule should be read at $\min_w G_{\text{eff}}(w)$ over the eigen directions of $[S^{-1}]_{PP}$. The two differ little in the designs run here. At the $G = 51$, $C = 5$, $m = 13$ design of Section 9, $G_{\text{eff}}(w)$ ranges from 35.92 to 36.42 over those directions, the size that (5.2) predicts for the nominal 0.05 chi-square test at $m = 13$ ranges from 0.378 to 0.385, and the largest m holding that predicted size below 0.075 is 2 at either end (`geff_eigen_sweep.csv`). Both G_{eff} and m are computable from the design alone, which turns Corollary 2 into an ex ante restriction on how many horizons a researcher may test.

Four objections to the blindness corollary A reader may object that we are restating a familiar fact, since post-treatment deviations from parallel trends are unrestricted by the data, so that any wedge switching on at treatment is absorbable and the equality of power and size follows at once. Rambachan and Roth (2023) make the same observation about their own identified set, writing that at any point in it "the null and alternative hypotheses are observationally equivalent, and so the most powerful test trivially has size α." We think the objection is correct about the mathematics and misses what Assumption 1 supplies. An unrestricted deviation is a free parameter about which nothing further can be said, whereas a recording response is a structural object whose sign is known from the direction of the recording change, whose magnitude is bounded by institutional facts about the recording

system, and which varies across outcome categories with the discretion the system has over each of them. The equivalence is what makes the deviation invisible, and the structure is what makes it recoverable from somewhere else. Neither half is of much use without the other.

Corollary 1 covers the test this paper goes on to propose. The exposure indices $d_1, \ldots, d_C$ are fixed known constants, so the cross-category statistic of Section 4 is a measurable function of the observed panel and the cohort labels, and its rejection probability is the same under both structures of Lemma 1. Nothing below separates a recording response from a latent deviation that happens to reproduce it. What changes between this section and the rest of the paper is the null under test rather than the information available. A pre-trend test asks whether the pre-treatment deviation is zero, and (3.2) makes it zero under both structures, which is why its power equals its size. The null (4.2) asks whether the post-treatment coefficients vary with the exposure index, which holds under measurement invariance and fails under any recording response whose path differs across categories. Both structures of Lemma 1 violate (4.2), and violate it identically. The cross section buys detection and not attribution. Reading a rejection as a recording response rather than as a latent deviation that tracks recording discretion is a judgment about which structures are plausible, and Assumption 1 is where that judgment is recorded.

That pre-treatment parallelism fails to guarantee post-treatment parallelism is long established (Kahn-Lang and Lang, 2020; Roth et al., 2023), and Ghanem et al. (2026) state that pre-trend tests are inherently uninformative where there are unrestricted structural breaks. A treatment-induced recording response is such a break. What we add is the mechanism that produces such a break, a construction showing that the induced deviation is exactly zero before treatment rather than merely unknown, and a statement about rejection probabilities rather than about informativeness. Corollary 1 is the version of that statement for one named mechanism, and the mechanism supplies what the general statement cannot: the deviation in (3.2) is not merely unbounded by pre-period behavior, it is identically zero there, by construction rather than by coincidence. A recording response is a structural break located exactly at the treatment date, which is the form Campbell and Stanley (1963) anticipated in words and which the results below put in a form that can be checked.

The construction resembles a familiar one and the resemblance is worth being precise about. Malani and Reif (2015) observe that anticipation effects and parallel trends violations are observationally equivalent, and Fenaroli (2026) works out the consequences for sensitivity analysis. The direction is what differs. An anticipation effect moves the pre period, which is why a pre-trend test can see it, and why the equivalence there is a problem about how to interpret a rejection. A recording response leaves the pre-period exactly as it was. There is nothing to interpret, because there is nothing to reject.

S3. ADDITIONAL APPLICATION EVIDENCE

The second panel is a state by year by condition panel of BRFSS crude prevalence covering 51 jurisdictions from 2011 to 2024, clustered on state. The treatment is a state's adoption of the ACA Medicaid expansion, with the cohort set to the calendar year of the coverage effective date when coverage began on or before July 1 and to the following year otherwise; ten states do not expand through 2024. Seven conditions carry the gradient, and the ex-

Table S3. Crime application: p values for $H_0 : \gamma_P = 0$ across nine designs and three loading normalizations

Design	Normalization	C	G_{eff}	χ^2	Hotelling	Perm.	Bootstrap	Sign
Baseline	reference	6	45.9	0.018	0.097	0.096	0.075	0.083
	naive			0.141	0.304	0.396	0.277	0.311
	triad			0.025	0.115	0.115	0.092	0.104
Pre 2020	reference	6	33.2	0.367	0.604	0.782	0.630	0.593
	naive			0.596	0.769	0.911	0.773	0.766
	triad			0.422	0.647	0.813	0.670	0.639
Drop rape	reference	5	45.9	0.016	0.089	0.075	0.063	0.048
	naive			0.008	0.062	0.067	0.040	0.052
	triad			0.018	0.096	0.083	0.066	0.051
Drop rape, larceny	reference	4	45.9	0.107	0.258	0.667	0.218	0.209
	naive			0.139	0.300	0.667	0.254	0.303
	triad			0.106	0.257	0.667	0.220	0.203
Cohort +1	reference	6	45.7	0.017	0.094	0.117	0.079	0.078
	naive			0.120	0.277	0.371	0.258	0.299
	triad			0.023	0.110	0.122	0.097	0.099
Threshold 0.8	reference	6	38.0	0.182	0.394	0.243	0.411	0.367
	naive			0.000	0.007	0.011	0.002	0.003
	triad			0.204	0.419	0.268	0.431	0.399
Core agency types	reference	6	46.9	0.012	0.074	0.032	0.062	0.062
	naive			0.164	0.328	0.389	0.306	0.359
	triad			0.018	0.093	0.051	0.083	0.091
Drop never treated	reference	6	45.3	0.017	0.094	0.172	0.093	0.094
	naive			0.079	0.220	0.338	0.216	0.245
	triad			0.026	0.119	0.207	0.121	0.131
Core types, pre 2020	reference	6	35.8	0.320	0.547	0.793	0.566	0.554
	naive			0.537	0.717	0.926	0.728	0.717
	triad			0.367	0.587	0.819	0.601	0.602

Notes: p values for the joint null $H_0 : \gamma_P = 0$ of (4.2) in the pooled gradient regression (4.1), estimated by OLS with the CR1 covariance clustered on state. Every design has $G = 51$ state clusters except drop never treated, which has $G = 47$, and every design tests $m = 9$ post-treatment restrictions. χ^2 is the χ^2_9 calibration; Hotelling is (5.2) evaluated at G_{eff}; Perm. is the label permutation test of Definition 2 enumerating all $C!$ assignments of δ, whose realized resolution floor is $1/720$ at $C = 6$, $1/120$ at $C = 5$ and $1/24$ at $C = 4$; Bootstrap is the restricted wild cluster bootstrap and Sign the cluster sign-change test, each with 1,999 draws. Naive sets every loading to one; triad is the S_C-equivariant estimator (6.1); adjusted fixes the reference category at the smallest d_c and is not covered by Theorem 1. The entry shown as 0.000 is 2.1×10^{-5}. Sources: `results/empirics/inference_gradient.csv` for the p values and `results/empirics/app_floors_injection.csv` for the realized floors.

posure index grades them by whether a clinical encounter is required for the respondent to report the condition: diagnosed diabetes 1.00, diagnosed high cholesterol 1.00, diagnosed hypertension 0.90, diagnosed depression 0.80, self reported obesity 0.05, current smoking 0.00 and no exercise in the past month 0.00. Three further series in the panel, diabetes mortality, insurance coverage and a past year checkup, are not used in the gradient.

Hypertension and high cholesterol are asked only in odd years, so Assumption 2 fails on the seven category stack. That failure costs the closed-form of Lemma 2 and not the conclusion of Theorem 1, which needs Assumption 4 in its place. What blocks exactness on the full set is that the cell pattern is a function of the category label, so Assumption 4 fails as well. The cluster level procedures of Section 7 use neither assumption and could be run on the unbalanced full set; we do not report that specification. We report two balanced sub designs: the five annually asked categories on all 709 state year cells, and all seven categories on the 353 odd-year cells. Under either equivariant normalization no category is dropped, so $C = 5$ and $C = 7$, with $G_{\text{eff}} = 36.0$ and 32.8, $\Sigma_\delta^2 = 0.958$ and 1.444, and support for $m = 2$ restrictions against the $m = 7$ tested. Neither sub design rejects (4.2) under the triad normalization, at chi square $p = 0.730$ and 0.391 and label permutation $p = 0.850$ and 0.854 over all $C!$ assignments, nor under the naive normalization, at chi square 0.295 and 0.967 and label permutation 0.867 and 0.964.

The health panel is also where the one-factor premise behind (6.1) is least credible. By the quantities of Section 6 the common factor carries 0.121 and 0.065 of the median category's demeaned variance in the two sub designs, against 0.397 in the crime panel, and the single triads disagree about ρ_c by factors of 2.39 and 8.59. The nonrejections above are therefore weak evidence in two separate senses. The design supports $m = 2$ restrictions and tests $m = 7$, and the normalization the equivariance of Theorem 1 runs through is estimating loadings on a factor that carries little of the variation. Neither point bears on the crime design, where the factor is the strongest of the three.

Every rejection anywhere in the health panels comes from the one normalization Theorem 1 does not cover, and none survives the correct reference set. The reference category rule fixes its reference at the position of the smallest d_c, which is not S_C-equivariant, and its loading guard drops diagnosed diabetes in the annual sub design and high cholesterol and no exercise in the odd-year sub design. Under it the odd-year sub design rejects at $p = 0.033$ on the level outcome and under every calibration on the log outcome. Where the guard fires, permuting only the survivors conditions on a multiset that is not ancillary: Assumption 4 makes the reference set the $C_0!/(C_0 - C)!$ ordered injections rather than the $C!$ permutations of the survivors, and recomputing against it moves the odd-year value from 0.033 to 0.062. Under the two equivariant normalizations nothing rejects on either outcome. Given Section 10 none of this is evidence about Medicaid expansion, and we report it because a normalization that is not equivariant is the failure mode Theorem 1 is designed to exclude and this is what it looks like in data.

The chi-square rejection is joint with the normalization. Under the naive normalization the chi-square test does not reject at all, at $p = 0.141$, and the label permutation returns $p = 0.396$; under the reference category normalization the chi-square test gives $p = 0.018$ and the label permutation $p = 0.096$. Table S3 reports all three normalizations for each of the nine crime designs, of which the design described above is one. The artifact that correct calibration overturns is therefore an artifact of the chi-square reference distribution applied

to a normalized outcome; under the naive normalization there is nothing to overturn.

A pre-period placebo in the same design behaves differently. Under the triad normalization its chi square p value is 6×10^{-6} and its label permutation p value over the same 720 assignments is 0.0167, the twelfth largest of the 720 orbit values. The exposure indices are not symmetric about zero and the realized orbit multiplicity is one, so Proposition 3(ii) does not apply and the floor is 1/720 rather than 2/720. Under the naive normalization the same permutation test returns $p = 0.153$, so the pre-period evidence, like the post-period evidence, is conditional on the normalization.

The two panels and how they are built The first design is an unusually direct instance of the mechanism of Section 3. A state's move to incident-based reporting changes the recording system itself, on a date the state chooses, so the recorded series can move without the offense rate moving at all. Lemma 1 says the pre-period cannot separate the two, and neither design below offers the announcement window that Section 3 identifies as the one place a placebo test retains power. The cross-category gradient is what is left.

The first panel is a state by year by offense panel of recorded index crime covering 51 states from 1979 to 2024, clustered on state. Recorded counts are the FBI's summary based estimated state totals and the outcome is the log offense rate per 100,000 residents. The treatment is a state's transition from summary reporting to the National Incident-Based Reporting System, dated as the first year in which at least half of the state's agencies on the FBI Crime Data Explorer roster carry a NIBRS start date; Florida, Hawaii, New York and Pennsylvania do not reach that share by 2024 and enter as never treated. Six offense categories carry the gradient, in ascending order of exposure: motor vehicle theft, aggravated assault, robbery, burglary, rape and larceny. Homicide is retained in the panel as a $d = 0$ series and is not used in the gradient. The exposure index is $d_c = 1 - r_c$, with r_c the share of victimizations reported to police averaged over the 2018 and 2019 National Crime Victimization Survey (Bureau of Justice Statistics, *Criminal Victimization 2019*, NCJ 255113, Table 6) and rescaled so the largest focal value is one, giving $d = (0.290, 0.604, 0.628, 0.699, 0.976, 1.000)$ in that order. The design has $G = 51$, $C = 6$, $m = 9$ and $\Sigma_\delta^2 = 0.349$, and the multiset $\{\delta_c\}$ is not symmetric about zero.

Each category array is divided by an estimated loading before the gradient is formed, because the categories respond by different amounts to the common latent component, and the choice of normalization determines whether Theorem 1 applies. We report three. The naive normalization sets every loading to one. The triad normalization is the log triad estimator (6.1) of Section 6, computed from the untreated cross-category covariance matrix and scaled to geometric mean one. Relabeling the categories relabels $\hat{\rho}$ identically, so by Lemma 3 the normalized arrays are an S_C-equivariant function of the raw arrays, the Wald statistic is order invariant, and Theorem 1 applies. The third normalization is the one the specification was originally run with, which fixes its reference category at the position of the smallest d_c. That choice makes the statistic depend on the order in which the pairs $(\delta_c, \mathcal{Y}_c)$ are listed, so Theorem 1 does not cover it, and we report it for comparison only.

Under the triad normalization the chi-square test of (4.2) rejects and none of the three randomization calibrations does. The design has $G_{\text{eff}} = 45.9$, which by the rule of Section 9 supports $m = 3$ restrictions at size 0.075; the specification tests $m = 9$, for which (5.2)

predicts a chi-square size of 0.169. The chi-square statistic is 19.00 with $p = 0.025$. The label permutation test of Definition 2 over all 720 assignments of δ returns $p = 0.115$, the restricted wild cluster bootstrap returns $p = 0.092$ and the cluster sign-change test returns $p = 0.104$. Referring W to (5.2) with G_{eff} in place of G returns $p = 0.115$; that substitution is the approximation of Definition 1 and not a calibration Proposition 2 licenses, and Section 9 prices its null rejection rate at 0.000 at $G = 20$ rising to 0.061 at $G = 100$, so it is reported here as a diagnostic. The realized resolution floor of the permutation test here is $1/720 = 0.0014$, so rejection at the 5% level was attainable and the nonrejection is not a consequence of Proposition 3.

The second panel is a state by year by condition panel of BRFSS crude prevalence covering 51 jurisdictions from 2011 to 2024, clustered on state, with seven conditions graded by whether a clinical encounter is needed for the respondent to report the condition, and treatment given by a state's adoption of the ACA Medicaid expansion. Section 10 screens it out before any test is run, at best ratios of 12.5 and 14.8 in the two balanced sub designs, and two further features of it are reported in the online Appendix: Assumption 2 fails on the seven category stack because hypertension and high cholesterol are asked only in odd years, and every rejection anywhere in these panels comes from the one normalization Theorem 1 does not cover and none survives the correct reference set.

The wedge each institution implies is what Section 10 compares the minimum detectable response with, and by (10.1) it is the projection of the per category response on the exposure index rather than the size of that response. For the crime panel the FBI and Bureau of Justice Statistics comparisons of the two reporting systems give per category log changes averaging 0.011, whose projections are 0.0044 and 0.0017 log points per unit d; the derivation is in `theory/institutional_wedge.md` in the replication archive. For the health panels the mechanism implies 0.052 percentage points for diagnosed diabetes, at the estimated 1.98 percentage point coverage gain and an undiagnosed share of 0.213. Both are magnitudes the recording institution makes plausible rather than estimates, and both are fixed before any test is run, which is what makes the comparison in Section 10 a design question rather than a reading of the outcome.

S4. ADDITIONAL IDENTIFICATION AND NORMALIZATION EVIDENCE

The three designs of Section 11 differ sharply on both. In the crime panel the factor carries 0.397 of the median category's demeaned variance, the largest eigenvalue of the cross-category correlation matrix carries 0.522 of its trace against the $1/C = 0.167$ that no factor at all would give, and single triads disagree about ρ_c by a factor of 1.23. In the annual and odd-year health sub designs the median shares are 0.121 and 0.065, the trace shares are 0.269 and 0.241 where no factor would give 0.200 and 0.143, and the single triads disagree by factors of 2.39 and 8.59. One category in the odd-year sub design has a share of 0.004. The premise is a serviceable approximation in the crime design and is not one in the health designs, where (6.1) averages quantities that do not agree with each other.

These are descriptive quantities and we attach no p value to them. A bootstrap reference fitted under a one-factor null is not calibrated for either: on data generated from a true one-factor structure at the dimensions of the three designs, with and without serial correlation

in the factor and in the idiosyncratic errors, a circular block bootstrap reference rejects between 0.090 and 0.995 on the eigenvalue ratio and between 0.090 and 0.178 on the triad dispersion, against a nominal 0.05 over 400 replications per cell. Fitting the null to the data biases both references, and we report the statistics rather than tests built on them. Sources: `onefactor_diagnostic.csv` and `onefactor_calibration.csv`.

What a failure of the premise costs is worth stating precisely. Lemma 3 is algebraic and holds whatever generates the data, and one-factor is not among the hypotheses of Theorem 1, which are Assumption 4 and order invariance. What the structure supplies is the reason to expect Assumption 4 to hold after normalizing, since dividing by an estimated loading equalizes the factor component of the arrays only when there is a common factor to equalize. Where the premise fails the conclusion of Theorem 1 is untouched and the case for its hypothesis is weaker, which is a reason to report the cluster level procedures of Section 7, since they use neither the normalization nor Assumption 4.

A second reading deserves an answer. Mikhaeil and Harshaw (2025) describe the same collapse and treat it as correct behavior, on the ground that an explicit conditional extrapolation assumption licenses inferring small post-treatment violations from small pre-treatment ones. That defence is sound where the assumption holds. A recording response is precisely a violation of it that passes every pre test by construction, so the assumption cannot be checked in the direction that matters here.

For the smoothness class the corresponding statement is weaker and older. That a researcher cannot calibrate M from the data and retain honest coverage is the nonadaptivity result of Low (1997) and Cai and Low (2004), stated for econometric practice by Armstrong and Kolesár (2018) and Armstrong and Kolesár (2020), who write that "to maintain honesty over the whole function class, a researcher must choose M a priori, rather than attempting to use a data driven method." Kolesár and Rothe (2018) put the asymmetry sharply: the data can bound the smoothness constant from below but never from above, and an honest interval needs an upper bound. Freeman (2026) restates the principle in a difference-in-differences framing. Nothing in this paper improves on any of that. What is new is the connection: $\Delta^{\mathrm{SD}}(M)$ is convex and centrosymmetric, so the nonadaptivity result applies to it directly, and Rambachan and Roth (2023) do not cite it. Their reference list contains Low (1995) for the computation of fixed length intervals and Armstrong and Kolesár (2018, 2020) for their optimality, but neither Low (1997) nor Cai and Low (2004).

The three conditions scoping the loading estimator Two conditions scope (6.1), and both restrict the procedure rather than the proof. First, L_{ck} discards the sign of $\hat{\Sigma}_{ck}$, so (6.1) estimates $|\theta_c|$: dividing a negatively loaded category by a positive $\hat{\rho}_c$ leaves its factor component with the wrong sign, and the exchangeability of Assumption 4 below then fails at every sample size. The maintained condition is that the loadings share a sign, which is checkable from the observed signs of the $\hat{\Sigma}_{ck}$. Second, the estimator is unstable at a loading near zero. At $\theta = (0.5, 0.8, 0.02, 1.4, 2.0)$ the largest error in $\hat{\rho}$ is 1.18 at 2,000 untreated cells and still 0.34 at 4×10^4, against 0.009 at 4×10^4 when the smallest loading is 0.5; the guard drops such a category rather than normalizing by it. The guard did not fire in any run reported here: over 400 replications in each of the twelve (G, C) cells of Section 9 the smallest $|\hat{\rho}_c|$ is 0.539. Sources: `verification.json` for the loading errors and

`loading_guard_check.csv` for the guard.

A third condition is the one-factor structure that (6.1) inverts, and unlike the first two it can be examined in the data. Two quantities computed from the same untreated covariance matrix bear on it. The first is the share of a category's demeaned variance that the common factor carries, $\hat{\theta}_c^2/\hat{\Sigma}_{cc}$, which is the quantity the normalization divides by. The second works in the units of (6.1) itself: for a category c and a pair $k < l$ drawn from the others, $(L_{ck} + L_{cl} - L_{kl})/2$ identifies $\log|\theta_c|$ up to a constant common to all c, and (6.1) averages those single triad values, so their dispersion within c measures the departure from one-factor where the estimator is sensitive to it.

In the three designs of Section 11 the common factor carries 0.397 of the median category's demeaned variance in the crime panel and 0.121 and 0.065 in the two health sub designs, and single triads disagree about ρ_c by factors of 1.23, 2.39 and 8.59. The premise is a serviceable approximation in the crime design and is not one in the health designs. The online Appendix gives the two statistics, a note on why no p value is attached to either, and the precise statement of what a failure costs: Lemma 3 is algebraic and survives, one-factor is not among the hypotheses of Theorem 1, and what goes is the reason to expect Assumption 4 to hold after normalizing.

S5. RELATED WORK AND THE APPLIED LITERATURE

Related work

Our exact null law is a statement about the cluster robust Wald statistic with G clusters, which places it in the few cluster inference literature. Cameron et al. (2008) propose the wild cluster bootstrap with the null imposed and document its size improvement over the cluster robust t; MacKinnon and Webb (2017) characterize when it fails. Neither paper treats the joint m restriction case in which m is comparable to G, and that is precisely where we find cross outcome falsification tests. Carter et al. (2017) show that cluster heterogeneity reduces the number of clusters that the asymptotics effectively deliver, and define the effective cluster count we use in Section 5. Canay et al. (2017) give asymptotic validity for a randomization test built on a group of fixed cardinality when the invariance holds only in the limit. The label permutation group of Section 6 has cardinality $C!$, which does not vary with G, so their framework covers designs in which the exchangeability of Assumption 4 holds for the category statistics only as G grows. The finite-sample argument itself is the classical group invariance construction, for which we follow Lehmann and Romano (2005).

We draw on four further results directly. MacKinnon et al. (2023) set out current practice for cluster-robust inference, and Section 7 takes the restricted wild cluster bootstrap in the form given there. Ibragimov and Müller (2010) obtain a t-test on cluster level estimates whose level is controlled with a small fixed number of heterogeneous clusters and no appeal to $G \to \infty$; that construction and the label permutation test of Section 6 differ in what is treated as the repeated unit, clusters there and outcome categories here. Young (2019) recomputes p values by randomization inference for a sample of published experimental papers in economics and reports that a substantial share of the rejections at conventional levels do not survive, which is the direction of the gap between the chi-square and the permutation p value in Section 11. Djogbenou et al. (2019) prove a central limit theorem for

the wild cluster bootstrap distribution conditional on the data, and that theorem supplies the asymptotic level of the cluster sign-change test of Section 7, which is the null imposed Rademacher wild cluster bootstrap written in score form.

We are not the first to correct both the variance estimator and the reference distribution for statistics of this kind, and a small-sample cluster-robust literature does both. Bester et al. (2011) hold the number of clusters fixed and let cluster size grow, and obtain limiting t and F distributions in which G plays the role of the sample size; the $F(m, G-m)$ shape of (5.2) is theirs as a large cluster approximation. Bell and McCaffrey (2002) remove the downward bias of the cluster-robust variance estimator under a working model and pair the correction with Satterthwaite degrees of freedom, and Pan and Wall (2002) do the same for generalized estimating equations. Imbens and Kolesár (2016) give degrees of freedom adjustments for the heteroskedasticity robust case. For a hypothesis with $m > 1$ constraints, Tipton and Pustejovsky (2015) refer the statistic to an approximate Hotelling T^2 whose denominator degrees of freedom are matched from the data, and Pustejovsky and Tipton (2018) extend that test to models with arbitrary sets of fixed effects and report Type I error close to the nominal level across the designs they simulate.

Proposition 2 does not introduce that reference distribution. It states an algebraic identity between the CR1 Wald statistic and a Hotelling T^2, and, under Assumption 3, an exact $F(m, G-m)$ law for the joint null that the post-treatment coefficients do not vary with exposure, holding at every G and m with $G - m \geq 1$ rather than at denominator degrees of freedom matched from the data, and rather than in the large cluster limit of Bester et al. (2011). What we add is the finite-sample statement and the explicit constant ω_2, rather than the family itself. Corollary 2 is the second difference. It gives the rate $m = o(G^{2/3})$ at which the chi-square calibration stays valid, the limit the size converges to when $m^{3/2}/G$ tends to a positive constant, and the convergence of the size to one when m/G does. Calhoun (2011) derives the corresponding correction for the regression F test when the number of restrictions grows with the number of observations rather than with the number of clusters, and reports the same direction of failure for the uncorrected test.

The design that motivates the problem is the use of graded outcome categories to detect a treatment-induced change in how outcomes are recorded. Weisburd et al. (2024) run this contrast across crime categories. Hausman et al. (1998) formalize outcome misclassification that depends on treatment status in a cross sectional discrete response setting, and Gutknecht and Liu (2024) study misclassified outcomes in a changes in changes design, and Denteh and Kédagni (2022) study a misclassified treatment indicator in difference-in-differences, a distinct channel: their designs maintain parallel trends and ask what misclassification does to the estimand, whereas an outcome recording response is itself a parallel trends violation. Freyaldenhoven et al. (2019) use an auxiliary series that loads on the same latent confound as the outcome and use it to purge event-study coefficients of that confound. The device is close to the normalization that Assumption 4 calls for here, and it is put to a different use: dividing each category outcome by an estimate of its loading on the common factor equalizes those loadings, which is what makes the category arrays exchangeable, and it removes nothing from the coefficient under test. We take the estimand and the identification argument from that literature as given. This paper is about the reference distribution.

Falsification tests are also run as pre-trend tests, where Roth (2022) shows that conditioning on passing one distorts the sampling distribution of the effect estimate, and Rambachan

and Roth (2023) replace the test with bounds on the post-treatment violation. Our own results concern the calibration of a cross outcome test rather than the consequences of conditioning on its outcome, and we note that the two problems compound, since a test with size 0.39 conditions on an event that is mostly noise.

The gap is a live one. Reviewing difference-in-differences designs for pandemic policy, Goodman-Bacon and Marcus (2020) warn that estimates "will be biased if testing improves differently in the treatment and control groups," then recommend event-study estimates, which "can reveal many of the biases outlined above." Corollary 1 says that when the policy is what changed the recording, that remedy rejects no more often than its nominal level. Richardson et al. (2025) report a formal pre-trend test as evidence that parallel trends holds, and separately concede that their estimated increase in reported intimate partner violence may reflect disclosure rather than incidence, without connecting the two. Baum et al. (2026) evaluate prenatal screening mandates, a treatment whose direct effect is on detection, and validate the design with a flat pre-period and a placebo in time test. Callaway and Li (2023) report that they cannot detect violations of parallel trends in pre-treatment periods even though those violations are probably present, and attribute the failure to pre tests being underpowered. Corollary 1 says that where the contamination is a recording response the failure is not a matter of power.

Where the threat is taken seriously, the response is already the method we propose here. Harasztosi and Lindner (2019) study a large Hungarian minimum wage increase and validate the design on the pre period, reporting that exposed and unexposed firms "follow a parallel trend before the minimum wage hike" and that the trend "broke exactly in 2001, when the minimum wage was raised." In an appendix they concede that "in the presence of tax evasion, some of the estimated effects may reflect only reporting behavior," and that firms absorbing the shock by declaring previously undeclared earnings would make "the change in reported wages overstate the actual wage change." They settle the question not from the pre-period but across outcomes: "if the main response to the minimum wage is simply reporting, it is not clear why firms would adjust their capital stock or raise their prices," and a reporting story predicts a negative effect on materials that the data do not show. That is a cross outcome falsification test conducted informally, and Corollary 1 explains why the pre-period could not have served instead. The rest of this paper supplies the reference distribution such a test needs when the number of clusters is small.

The sharpest illustration is a design that works. Studying upcoding in Medicare Advantage, Geruso and Layton (2020) observe that "because MA risk scores are calculated using diagnosis data from the prior plan year, changes in MA enrollment can plausibly affect reported risk scores via differential coding only with a lag. Thus, contemporaneous penetration acts as a placebo test." Their placebo has power because an institutional accident supplies a window in which the coding response cannot yet have operated, and the identifying content is that window rather than the pre-period as such. Corollary 1 is the statement of what is left when no such window exists.

The full discussion of the closest prior work

S6. RELATED WORK

Our exact null law concerns the cluster robust Wald statistic, which places it in the few cluster literature. Cameron et al. (2008) propose the wild cluster bootstrap with the null imposed and MacKinnon and Webb (2017) characterize when it fails; neither treats the joint case in which the number of restrictions is comparable to the number of clusters, which is where cross outcome falsification tests live. MacKinnon et al. (2023) survey what is known. Carter et al. (2017) define the effective cluster count used in Section 5. Canay et al. (2017) give asymptotic validity for a randomization test on a group of fixed cardinality; the label permutation group of Section 6 has cardinality $C!$, which does not vary with the number of clusters. The finite-sample argument is the classical group invariance construction of Lehmann and Romano (2005).

We are not the first to correct both the variance estimator and the reference distribution. Bester et al. (2011) hold the number of clusters fixed and let cluster size grow, and the $F(m, G - m)$ shape of (5.2) is theirs as a large cluster approximation. Bell and McCaffrey (2002) remove the downward bias of the cluster-robust variance estimator and pair the correction with Satterthwaite degrees of freedom, and Imbens and Kolesár (2016) give degrees of freedom adjustments for the heteroskedasticity robust case. For a hypothesis with several constraints Tipton and Pustejovsky (2015) and Pustejovsky and Tipton (2018) refer the statistic to an approximate Hotelling law whose denominator degrees of freedom are matched from the data. Proposition 2 does not introduce that family. It states an algebraic identity and, under Assumption 3, an exact law holding at every cluster count and every number of restrictions with $G - m \geq 1$, rather than at matched degrees of freedom or in a large cluster limit. What we add is the finite-sample statement and the explicit constant ω_2. Corollary 2 adds the rate at which the chi-square calibration stays valid.

The design that motivates the problem is the use of graded outcome categories to detect a treatment-induced change in recording. Weisburd et al. (2024) run this contrast across crime categories. Hausman et al. (1998) formalize outcome misclassification in a cross section, and Freyaldenhoven et al. (2019) use an auxiliary series loading on the same latent confound to purge event-study coefficients of it, a device close to the normalization Assumption 4 calls for and put to a different use. Falsification tests are also run as pre-trend tests, where Roth (2022) shows that conditioning on passing one distorts the sampling distribution of the effect estimate and Rambachan and Roth (2023) replace the test with bounds. The online Appendix gives the extended discussion. This paper is about the reference distribution.

S7. THREE FINDINGS RESOLVED BY REPLICATION

Three findings an earlier draft of Section 9 reported without a cause are resolved by replication, and the online Appendix gives the working. The decline in measured chi-square size across the number of categories does not reproduce in an independent run. The nonmonotone power at five categories is a ceiling set by a reference set of 120 points rather than a turning point, the realized signal being monotone throughout. The Kolmogorov Smirnov rejection in the ideal experiment is what a correctly sized test does one time in twenty, at rates of 0.060, 0.045, 0.035 and 0.060 over 200 independent trials.

Three findings in Section 9 that an earlier draft reported without a cause are resolved

by replication. The measured chi-square size falls from 0.393 at $C = 3$ to 0.356 at $C = 8$ at $G = 51$, a decline of 2.45 standard errors, where (5.2) predicts almost no movement, and the draft attributed it to the loading normalization. An independent run of the same twelve cells at 2,000 draws gives 0.388, 0.385 and 0.384, flat to within a quarter of a standard error, and removing the normalization entirely gives 0.396, 0.391 and 0.396, also flat. The decline is neither the normalization nor a feature of the design.

The size-adjusted power of the two-sided label permutation test at $C = 5$ rises to 0.452 at $\kappa = 0.60$ and falls to 0.383 at $\kappa = 0.80$ in Table 3, a difference of 2.4 standard errors. Two candidates separate. The realized signal is strictly monotone: over 600 replications per cell the mean post-treatment gradient rises through 0.141, 0.267, 0.374, 0.450 and 0.488 and the median Wald statistic through 24.7, 36.8, 56.1, 70.2 and 79.0 as κ rises from 0.15 to 0.80, so neither the statistic nor the log wedge turns over. Size adjusted power over the same grid runs 0.093, 0.193, 0.335, 0.335 and 0.348, which is a plateau and not a decline. What the permutation test has at $C = 5$ is a ceiling rather than a turning point: the reference set has 120 points, and once the observed statistic is larger than most of the relabeled ones a further increase in κ cannot raise its rank. The large-sample local power calculation, which knows nothing of the reference set, is the same statement from the other side, and it is why it overstates simulated power by between 15 and 54 percentage points here.

The Kolmogorov Smirnov test of p value uniformity rejects at $G = 30$ in the experiment built to satisfy Assumption 3. Repeating that test on 200 independent sets of 1,000 draws gives rejection rates of 0.060, 0.045, 0.035 and 0.060 at $G = 20$, 30, 51 and 100, each within one Monte Carlo standard error of 0.015 of the nominal 0.05. The reported rejection is what a correctly sized test does one time in twenty. Source: `anomalies.csv`.

S8. THE EXACT SIZE AGAINST THE NUMBER OF RESTRICTIONS

S9. STAGGERED ADOPTION

Proposition 1 is stated for a single adoption cohort. The construction behind it is a statement about one cohort's pre period, and with several cohorts the comparisons that identify a later cohort before it adopts can involve units that have already adopted and therefore already carry the recording wedge. If that channel leaves a pre-period signature, a pre-period calibration can see the recording response after all and the blindness result is weaker under staggering than Table S1 suggests.

Table S4 settles the question by simulation. It repeats the experiment of Table S1 with one change: three cohorts adopt at periods 6, 8 and 10 with 40 percent of units never treated, in place of one cohort adopting at period 8 with half the units never treated. The estimator, the reference implementation of the honest sets, the calibration of the smoothness bound, the latent effect, the recording wedge and the target horizon are all held fixed, so the two tables are comparable cell for cell.

The collapse is the same and arrives sooner. Every relative magnitudes column reaches zero coverage of the latent effect one sample size earlier than it does in Table S1, the smoothness class calibrated from the pre-period fails at $N = 102{,}400$ rather than at $N = 409{,}600$, and the bound set from the recording institution holds at 1.000 throughout. The calibrated bound $\widehat{M}$ is smaller at every sample size than its single cohort counterpart, which is why the

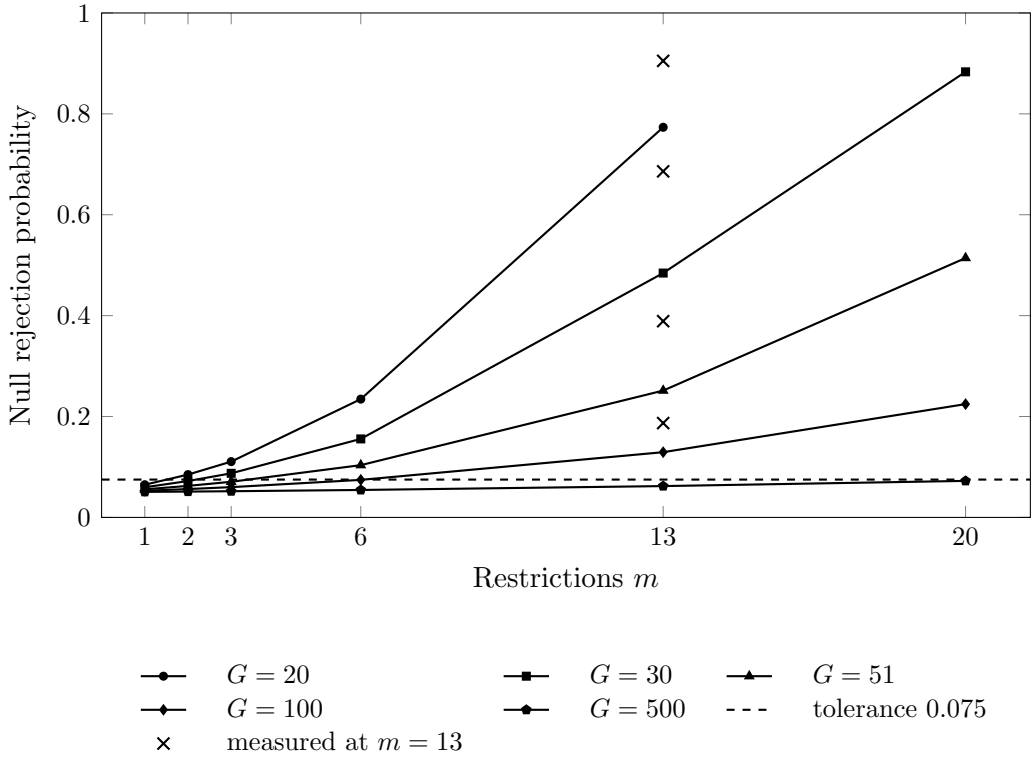


Figure S1. Exact size of the nominal 5% chi-square test against the number of restrictions

Notes: each curve gives the rejection probability of the nominal level-0.05 chi-square test of the joint null $H_0 : \gamma_P = 0$ in (4.2), computed from (5.2) under Assumptions 2 and 3 at $\omega_2 = 1$, against the number of restrictions m, at five cluster counts G. The dashed line marks a tolerance of 0.075. The crosses are the measured rejection rates at $m = 13$ and $C = 5$ from Table 2, at $G = 20$, 30, 51 and 100 reading down, over 2,000 replications each with a Monte Carlo standard error of at most 0.011. They lie above the corresponding curves because the simulated design gives G_{eff} between $0.7142\,G$ and $0.7273\,G$ rather than the equal cluster weights of Assumption 3(ii). Sources: `results/inference/anatomy_expansion.csv`, `results/inference/size_table.csv` and `results/inference/anatomy_effective_clusters.csv`.

crossing happens earlier: a smaller bound admits a narrower set, and the set has to contain a latent effect the recorded coefficients are displaced away from.

The fourth column is what rules the contamination channel out. If the recording response reached the pre-period through already-treated comparison units, the largest pre-treatment coefficient would settle at a nonzero constant. It does not. Multiplying it by $\sqrt{N}$ gives 3.081, 2.908, 3.082, 2.992 and 3.034 down the five rows, flat across a 256 fold range of the panel size, so the pre-period carries sampling noise and nothing else. At the largest panel the recording wedge is 63 times the largest pre-treatment coefficient it produces.

This is evidence rather than proof, and Proposition 1 remains stated for one cohort. What the table removes is the possibility that staggering is what rescues the pre period, which is the reading a reader of the scope restriction might otherwise reach.

Table S4. Coverage of the latent effect under a recording response, staggered adoption. Three cohorts adopting at periods 6, 8 and 10, 40 percent never treated, latent effect 1.0, recording wedge 0.6 at the target horizon, nominal 95%. The column $\max|\hat{\beta}_{\text{pre}}|$ is the mean over replications of the largest pre-treatment event-study coefficient in absolute value. Everything else follows Table S1. Source: `results/coverage/coverage_staggered.json`, produced by `scripts/oa_staggered.py`.

				$\Delta^{\text{RM}}(\bar{M})$			$\Delta^{\text{SD}}(M)$	
N	reps	$\widehat{M}$	$\max\|\hat{\beta}_{\text{pre}}\|$	$\bar{M}=0.5$	$\bar{M}=1$	$\bar{M}=2$	$M=\widehat{M}$	institutional
400	200	0.5376	0.1541	0.300	0.735	0.955	1.000	1.000
1,600	200	0.2729	0.0727	0.000	0.150	0.645	1.000	1.000
6,400	200	0.1370	0.0385	0.000	0.000	0.085	1.000	1.000
25,600	100	0.0653	0.0187	0.000	0.000	0.000	0.880	1.000
102,400	50	0.0344	0.0095	0.000	0.000	0.000	0.040	1.000

REFERENCES

Armstrong, T. B., & Kolesár, M. (2018). Optimal inference in a class of regression models. *Econometrica*, 86, 655–683. https://doi.org/10.3982/ecta14434

Armstrong, T. B., & Kolesár, M. (2020). Simple and honest confidence intervals in nonparametric regression. *Quantitative Economics*, 11, 1–39. https://doi.org/10.3982/qe1199

Arora, P., & Chand, A. (2026). Trading scope for credibility in difference-in-differences. arXiv. https://arxiv.org/abs/2608.16867

Baum, A., Agha, L., Menzies, N., & Cohen, A. (2026). Prenatal syphilis screening mandates and maternal syphilis case detection. *JAMA Health Forum*, 7, e260123. https://doi.org/10.1001/jamahealthforum.2026.0123

Bell, R. M., & McCaffrey, D. F. (2002). Bias Reduction in standard errors for linear regression with multi-stage samples. *Survey Methodology*, 28, 169–181.

Bester, C. A., Conley, T. G., & Hansen, C. B. (2011). Inference with dependent data using cluster covariance estimators. *Journal of Econometrics*, 165, 137–151. https://doi.org/10.1016/j.jeconom.2011.01.007

Cai, T. T., & Low, M. G. (2004). An adaptation theory for nonparametric confidence intervals. *The Annals of Statistics*, 32, 1805–1840. https://doi.org/10.1214/009053604000000049

Calhoun, G. (2011). Hypothesis testing in linear regression when k/n is large. *Journal of Econometrics*, 165, 163–174. https://doi.org/10.1016/j.jeconom.2011.07.003

Callaway, B., & Li, T. (2023). Policy evaluation during a pandemic. *Journal of Econometrics*, 236, 105454. https://doi.org/10.1016/j.jeconom.2023.03.009

Cameron, A. C., Gelbach, J. B., & Miller, D. L. (2008). Bootstrap-based improvements for inference with clustered errors. *Review of Economics and Statistics*, 90, 414–427. https://doi.org/10.1162/rest.90.3.414

Campbell, D. T., & Stanley, J. C. (1963). *Experimental and quasi-experimental designs for research*. Rand McNally.

Canay, I. A., Romano, J. P., & Shaikh, A. M. (2017). Randomization tests under an approximate symmetry assumption. *Econometrica*, 85, 1013–1030. https://doi.org/10.3982/ecta13081

Carter, A. V., Schnepel, K. T., & Steigerwald, D. G. (2017). Asymptotic behavior of a t-test robust to cluster heterogeneity. *Review of Economics and Statistics*, 99, 698–709. https://doi.org/10.1162/rest_a_00639

Denteh, A., & Kédagni, D. (2022). Misclassification in difference-in-differences models. arXiv. https://arxiv.org/abs/2207.11890

DiCiccio, C. J., & Romano, J. P. (2017). Robust permutation tests for correlation and regression coefficients. *Journal of the American Statistical Association*, 112, 1211–1220. https://doi.org/10.1080/01621459.2016.1202117

Djogbenou, A. A., MacKinnon, J. G., & Nielsen, M. Ø. (2019). Asymptotic theory and wild bootstrap inference with clustered errors. *Journal of Econometrics*, 212, 393–412. https://doi.org/10.1016/j.jeconom.2019.04.035

Fenaroli, F. (2026). A joint analysis of sensitivity to anticipation and parallel trends violations. arXiv. https://arxiv.org/abs/2603.00868

Freyaldenhoven, S., Hansen, C., & Shapiro, J. M. (2019). Pre-event trends in the panel event-study design. *American Economic Review*, 109, 3307–3338. https://doi.org/10.1257/aer.20180609

Freeman, H. (2026). Local asymptotic power of honest confidence intervals. arXiv. https://arxiv.org/abs/2607.10558

Geruso, M., & Layton, T. (2020). Upcoding: Evidence from Medicare on squishy risk adjustment. *Journal of Political Economy*, 128, 984–1026. https://doi.org/10.1086/704756

Ghanem, D., Sant'Anna, P. H. C., & Wüthrich, K. (2026). When should pre-trends be parallel? *AEA Papers and Proceedings*. https://doi.org/10.1257/pandp.20261109

Goodman-Bacon, A., & Marcus, J. (2020). Using difference-in-differences to identify causal effects of COVID-19 policies. *Survey Research Methods*, 14, 153–158.

Gutknecht, D., & Liu, C. (2024). Changes-in-changes for ordered choice models with under-reporting. arXiv. https://arxiv.org/abs/2401.00618

Harasztosi, P., & Lindner, A. (2019). Who pays for the minimum wage? *American Economic Review*, 109, 2693–2727. https://doi.org/10.1257/aer.20171445

Hausman, J. A., Abrevaya, J., & Scott-Morton, F. M. (1998). Misclassification of the dependent variable in a discrete-response setting. *Journal of Econometrics*, 87, 239–269. https://doi.org/10.1016/s0304-4076(98)00015-3

Hotelling, H. (1931). The generalization of Student's ratio. *Annals of Mathematical Statistics*, 2, 360–378.

Ibragimov, R., & Müller, U. K. (2010). t-statistic based correlation and heterogeneity robust inference. *Journal of Business & Economic Statistics*, 28, 453–468. https://doi.org/10.1198/jbes.2009.08046

Imbens, G. W., & Kolesár, M. (2016). Robust standard errors in small samples: Some practical advice. *Review of Economics and Statistics*, 98, 701–712. https://doi.org/10.1162/rest_a_00552

Kahn-Lang, A., & Lang, K. (2020). The promise and pitfalls of differences-in-differences: Reflections on 16 and pregnant and other applications. *Journal of Business & Economic Statistics*, 38, 613–620. https://doi.org/10.1080/07350015.2018.1546591

Kolesár, M., & Rothe, C. (2018). Inference in regression discontinuity designs with a discrete running variable. *American Economic Review*, 108, 2277–2304. https://doi.org/10.1257/aer.20160945

Kudô, A. (1963). A multivariate analogue of the one-sided test. *Biometrika*, 50(3–4), 403–418. https://doi.org/10.2307/2333909

Lehmann, E. L., & Romano, J. P. (2005). *Testing statistical hypotheses* (3rd ed.). Springer.

Low, M. G. (1995). Bias-variance tradeoffs in functional estimation problems. *The Annals of Statistics*, 23, 824–835. https://doi.org/10.1214/aos/1176324624

Low, M. G. (1997). On nonparametric confidence intervals. *The Annals of Statistics*, 25, 2547–2554.

MacKinnon, J. G., & Webb, M. D. (2017). Wild bootstrap inference for wildly different cluster sizes. *Journal of Applied Econometrics*, 32, 233–254. https://doi.org/10.1002/jae.2508

MacKinnon, J. G., M. Ø. Nielsen, & Webb, M. D. (2023). Cluster-robust inference: A guide to empirical practice. *Journal of Econometrics*, 232, 272–299. https://doi.org/10.1016/j.jeconom.2022.04.001

Malani, A., & Reif, J. (2015). Interpreting pre-trends as anticipation: Impact on estimated treatment effects from tort reform. *Journal of Public Economics*, 124, 1–17. https://doi.org/10.1016/j.jpubeco.2015.01.001

Mikhaeil, J. M., & Harshaw, C. (2025). Valid inference when testing violations of parallel trends for difference-in-differences. arXiv. https://arxiv.org/abs/2510.26470

Pan, W., & Wall, M. M. (2002). Small-sample adjustments in using the sandwich variance estimator in generalized estimating equations. *Statistics in Medicine*, 21, 1429–1441. https://doi.org/10.1002/sim.1142

Pustejovsky, J. E., & Tipton, E. (2018). Small-sample methods for cluster-robust variance estimation and hypothesis testing in fixed effects models. *Journal of Business & Economic Statistics*, 36, 672–683. https://doi.org/10.1080/07350015.2016.1247004

Rambachan, A., & Roth, J. (2023). A more credible approach to parallel trends. *Review of Economic Studies*, 90, 2555–2591. https://doi.org/10.1093/restud/rdad018

Richardson, R., Raj, A., Downs, J., Bergenfeld, N., Wiltshire, R., Dimitrova, A., & Benmarhnia, T. (2025). Impact of domestic violence legislation on reported intimate partner violence: A difference-in-differences analysis across 16 countries. *SSM–Population Health*, 43, 101837. https://doi.org/10.1016/j.ssmph.2025.101837

Roth, J. (2022). Pretest with caution: Event-study estimates after testing for parallel trends. *American Economic Review: Insights*, 4, 305–322. https://doi.org/10.1257/aeri.20210236

Roth, J., Sant'Anna, P. H. C., Bilinski, A., & Poe, J. (2023). What's trending in difference-in-differences? a synthesis of the recent econometrics literature. *Journal of Econometrics*, 235, 2218–2244. https://doi.org/10.1016/j.jeconom.2023.03.008

Shapiro, A. (1988). Towards a unified theory of inequality constrained testing in multivariate analysis. *International Statistical Review*, 56, 49–62.

Tipton, E., & Pustejovsky, J. E. (2015). Small-sample adjustments for tests of moderators and model fit using robust variance estimation in meta-regression. *Journal of Educational and Behavioral Statistics*, 40, 604–634. https://doi.org/10.3102/1076998615606099

Weisburd, D., Wilson, D. B., Gill, C., Kuen, K., & Zastrow, T. (2024). Are we underestimating the Crime prevention outcomes of community policing? the importance of Crime reporting sensitivity bias. *Journal of Law and Empirical Analysis*, 1, 1–17. https://doi.org/10.1177/2755323x241233469

Young, A. (2019). Channeling Fisher: Randomization tests and the statistical insignificance of seemingly significant experimental results. *Quarterly Journal of Economics*, 134, 557–598. https://doi.org/10.1093/qje/qjy029